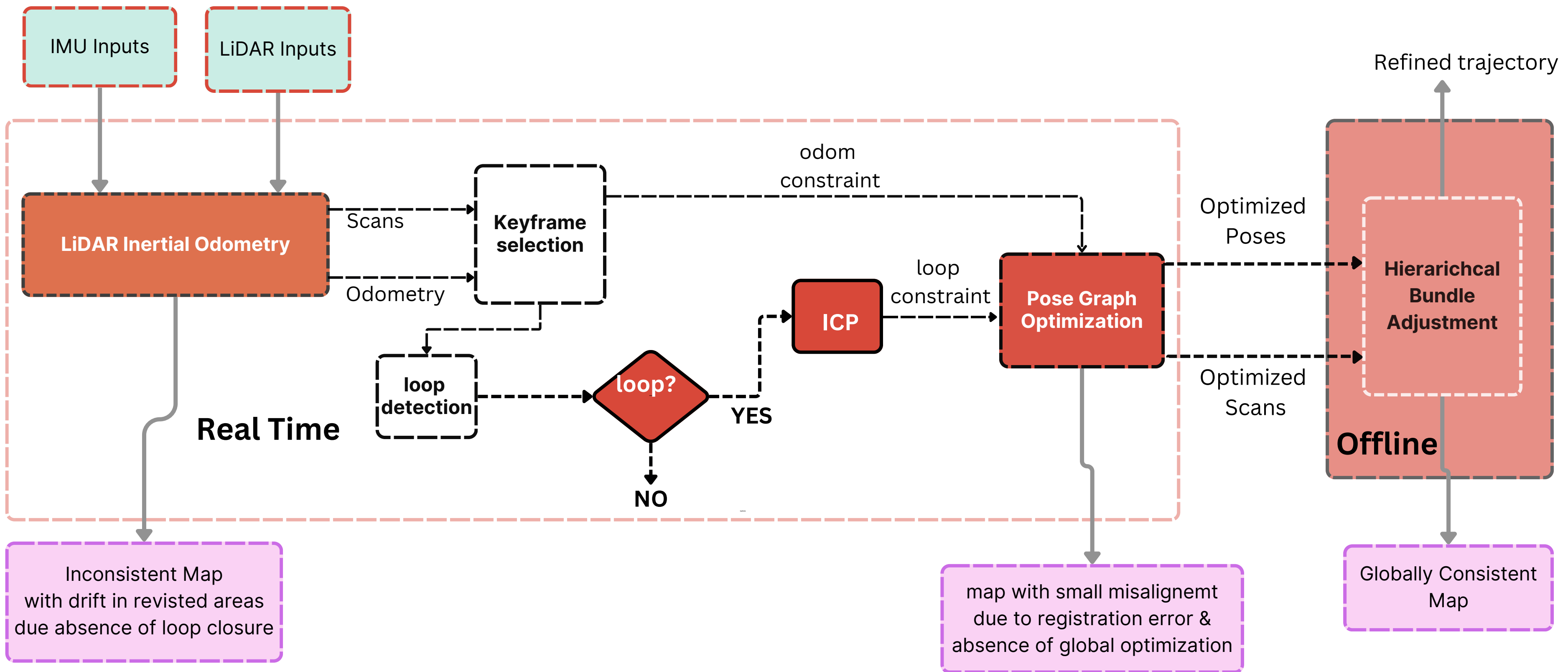




CONTENT

01

PROGRESS





SAXION
UNIVERSITY OF

Smart Mechatronics
and Robotics



SENSORS

- IMU -
- LIDAR - Ouster-128
- Camera - Z-camera(RGPD)



Problem Statment

High-precision and robust localization in Indoor and outdoor

State of Art Solutions

01

GNSS

• Challenges

- Failed in GPS denied area
- Expensive

02

SLAM

• Challenges

- Drift in long time run with out loop closure



Map Based Localization

01

Generate Accurate Map of The Env't

- Could be used GNSS or SLAM

02

Localize inside map

- Estimate the pose of the robot by scan matching with map
- Fuse with LIDAR Odometry or Wheel Odometer using sensor fusion



Scan to Map matching

01

ICP

Iteratively refines the alignment by minimizing the distance between corresponding points in the scan and the map.

Advantage

- Simple, widely used, and effective in structured environments
- Different Variant could be used for different scenario

Challenge

- More Sensitive to initial pose estimates and outliers
- Can be trape to Local minimum



Scan to Map matching

02

NDT

- Represents the map as a grid of normal distributions instead of discrete points then matches scans to distributions rather than points

Advantage

- More Robust and Higher Convergence
- Matches scans to distributions rather than points

Challenge

- could be slower than ICP and sensitive to parameter
- Sensitive to initial pose estimates



Scan to Map matching

03

Deep Learning based

These approaches leverage neural networks to improve feature extraction and matching

Advantage

- Handles complex scenarios
- Robust to Env't change

Challenge

- Requires large, diverse datasets for training.
- Computationally intensive, often needing GPUs
- Susceptible to overfitting.



Challenges

01

Dependence on High-Definition (HD) Maps

- small error in map quality can cause significant localization error

02

Environmental Dynamics

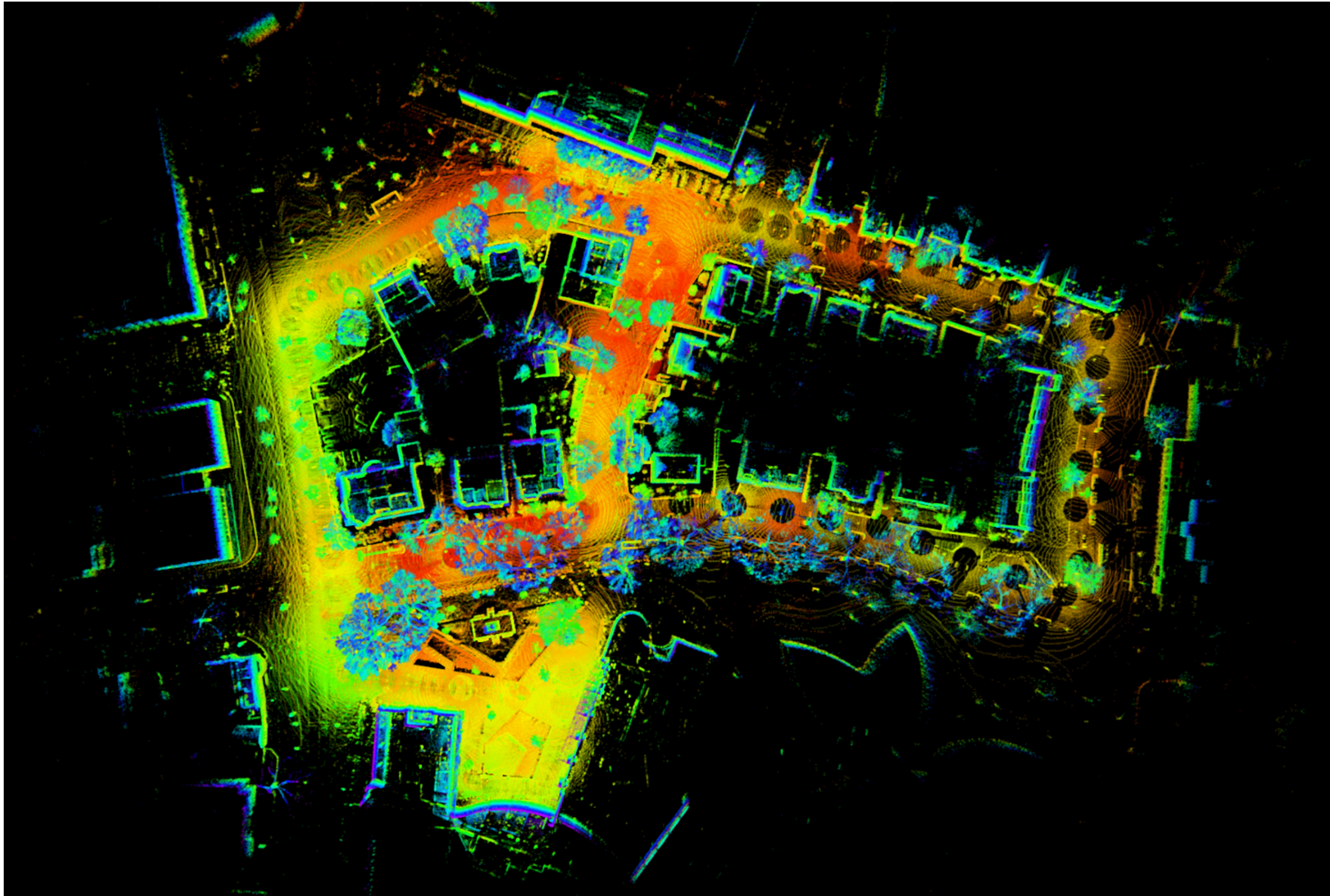
- Dynamic object and bad weather condition
- 💡 remove cars ,pedestrian from pointcloud

03

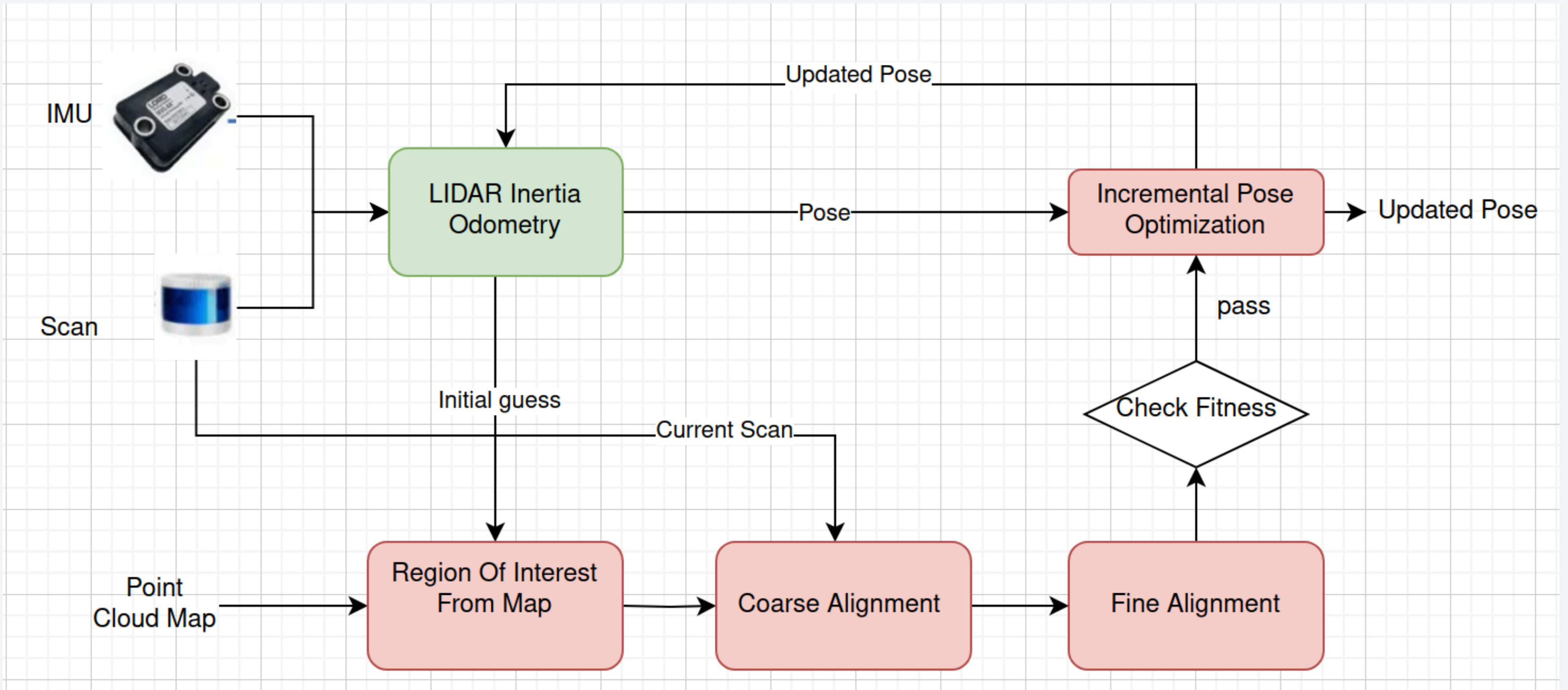
Environmental Computational and Real-Time Constraints

- to process map , scan to map matching
- 💡 Create Local map on the field of viiew of the current pose

Where I am in the map ?



METHODOLOGY DIAGRAM



SUB-MAP RETRIEVAL

- minimize the Field of view of lidar(Flexible)
- Cut the global map based on the FOV of lidar

ACCURACY OF REGISTRATION

PCL LIBRARY IMPLEMENTATION

- Using The PCL library implementation of registration - ICP , GICP , NDT
- ICP Accuracy - fine
- ICP Time - 0.6 seconds
- GICP and NDT Time - more than 1 seconds

ACCURACY OF REGISTRATION

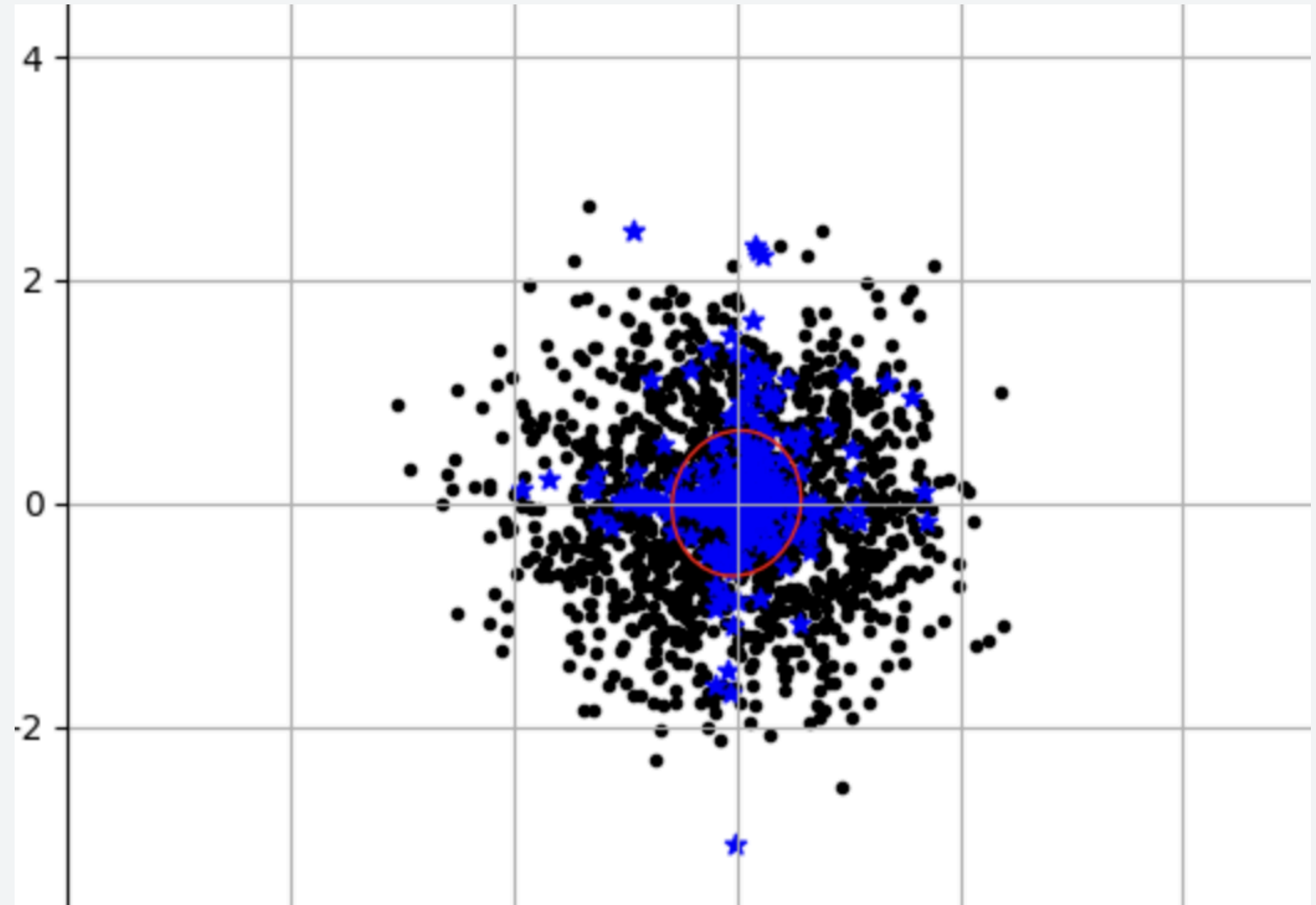
CUDA IMPLEMENTATION

- Optimized for real time usage
- Accuracy - fine
- CUDA ICP Time - 0.078 seconds Avarage
- CUDA GICP - 0.078 seconds Avarage

PCL ICP

- Average Fitness - 0.13
- Time - 0.43 seconds Avarage
- Std 0.5 Noise to X,Y,Z
- Std 3 degree Noise to Yaw Angle

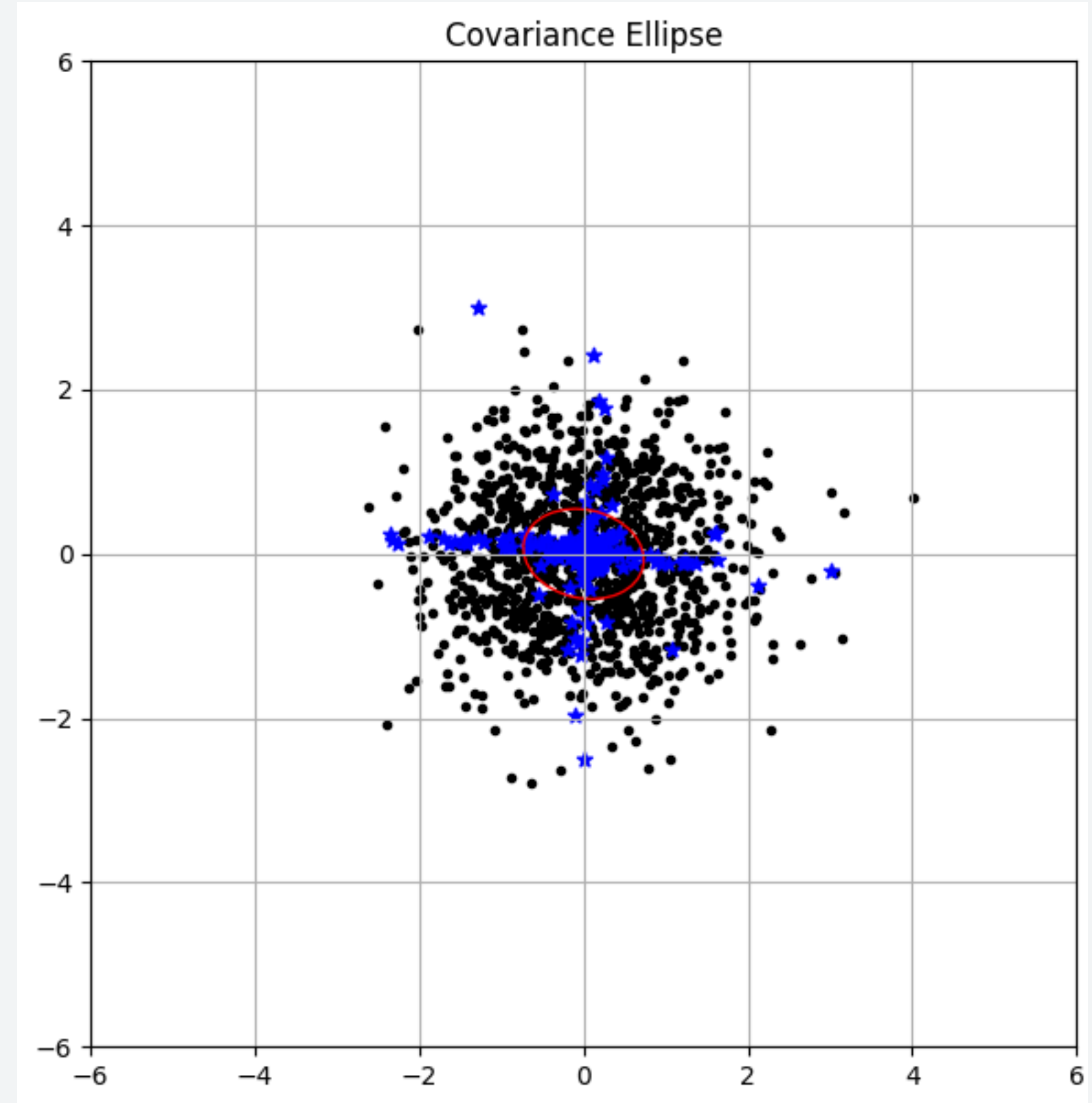
```
[ [0.05589435 0.00381775]  
  [0.00381775 0.07087967]]
```



CUDA ICP

- Average Fitness - 0.13
- Time - 0.078 seconds Avarage
- Std 0.5 Noise to X,Y,Z
- Std 3 degree Noise to Yaw
Angle

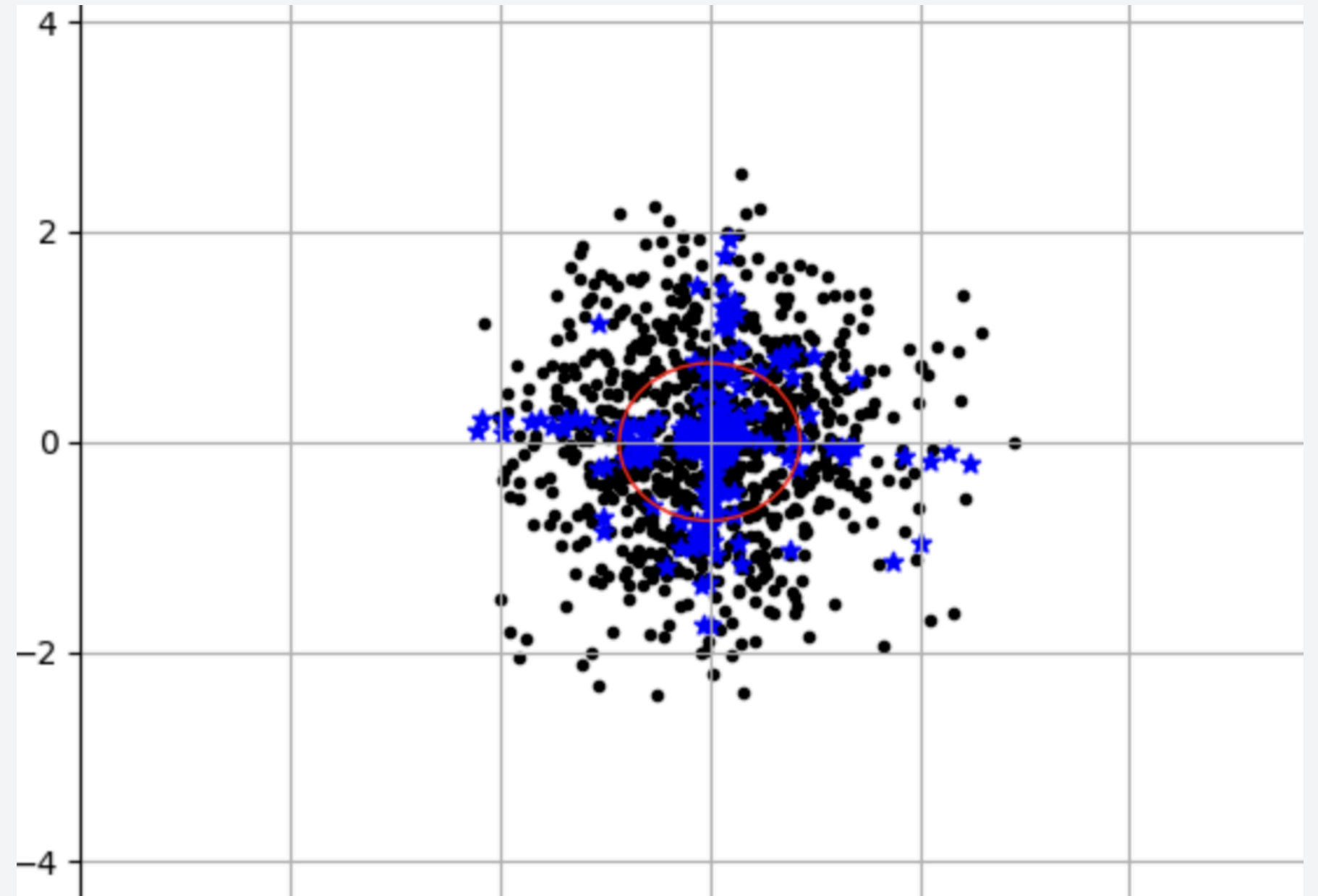
```
[[ 0.08951205 -0.00890963]  
 [-0.00890963  0.05016512]]
```



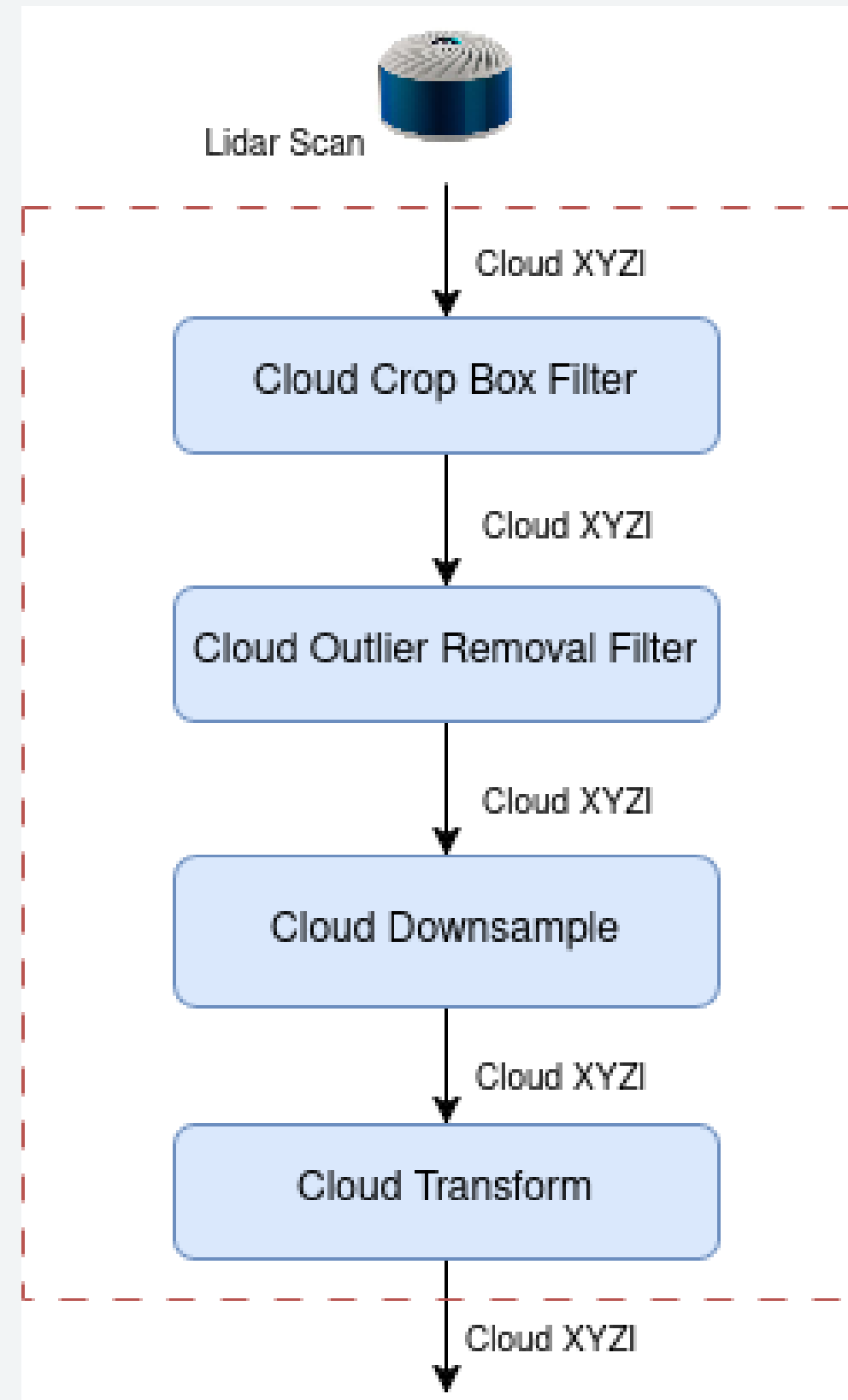
CUDA ICP

- Average Fitness - 0.18
- Time - 0.083 seconds Average
- Std 1 Meter Noise to X,Y,Z
- Std 10 degree Noise to Yaw Angle

```
[[0.12393783 0.00097468]  
 [0.00097468 0.09398669]]
```

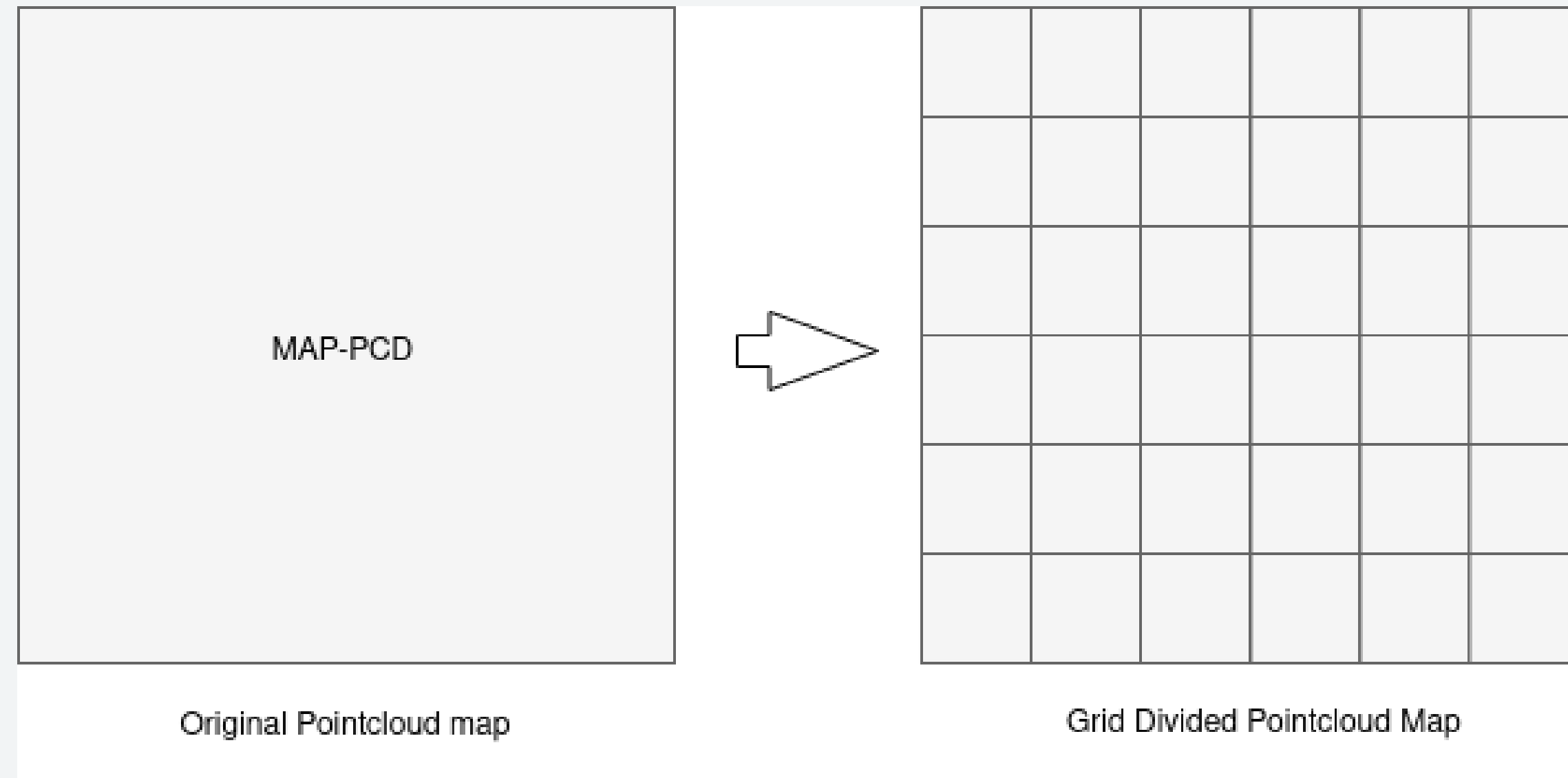


LIDAR SCAN PREPROCESSING

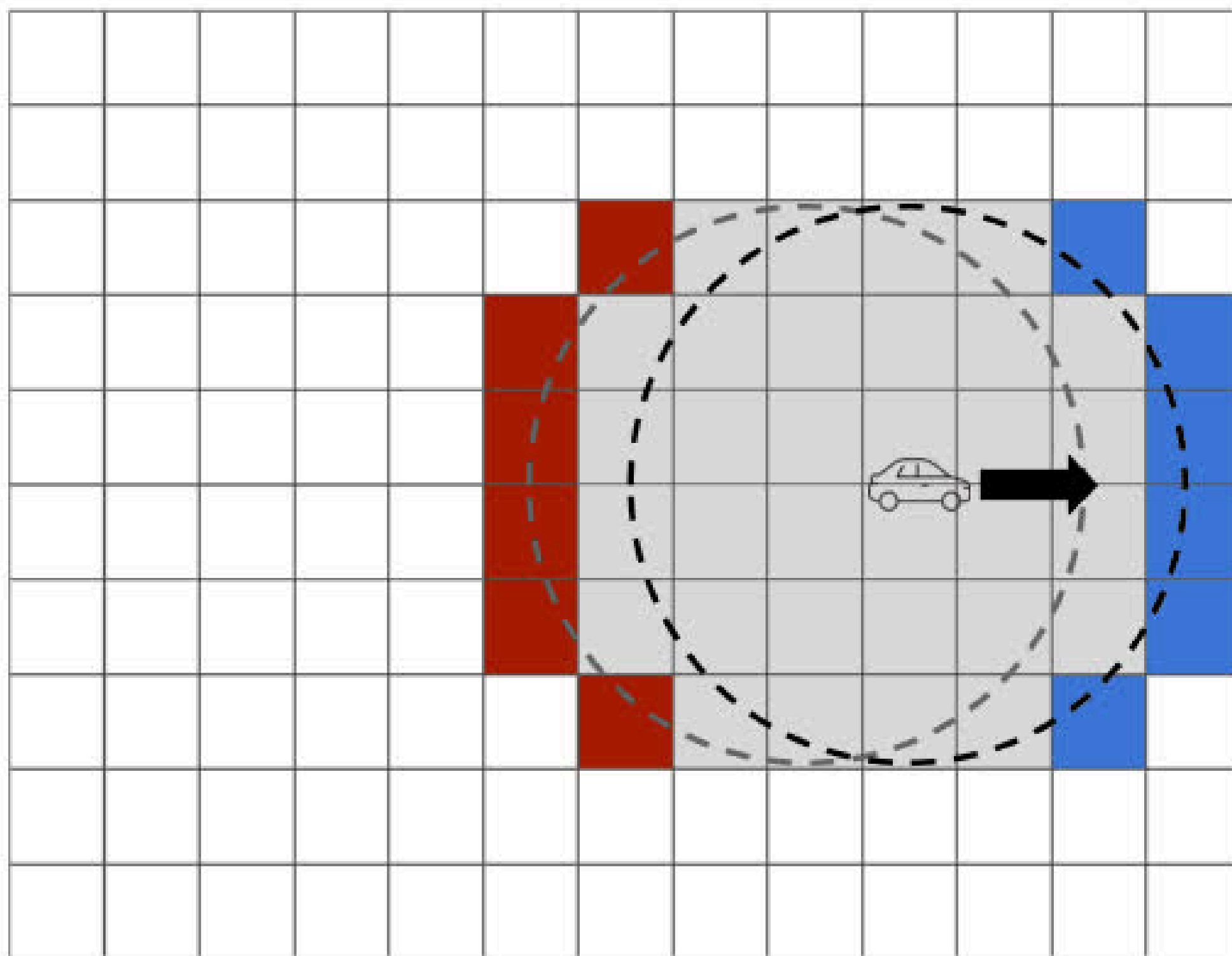


MAP PROCESSING

- Divide the map to gride
- Assign Unique Index to each gride



LOCAL MAP LOADING(RETRIVAL)



$t = 3$

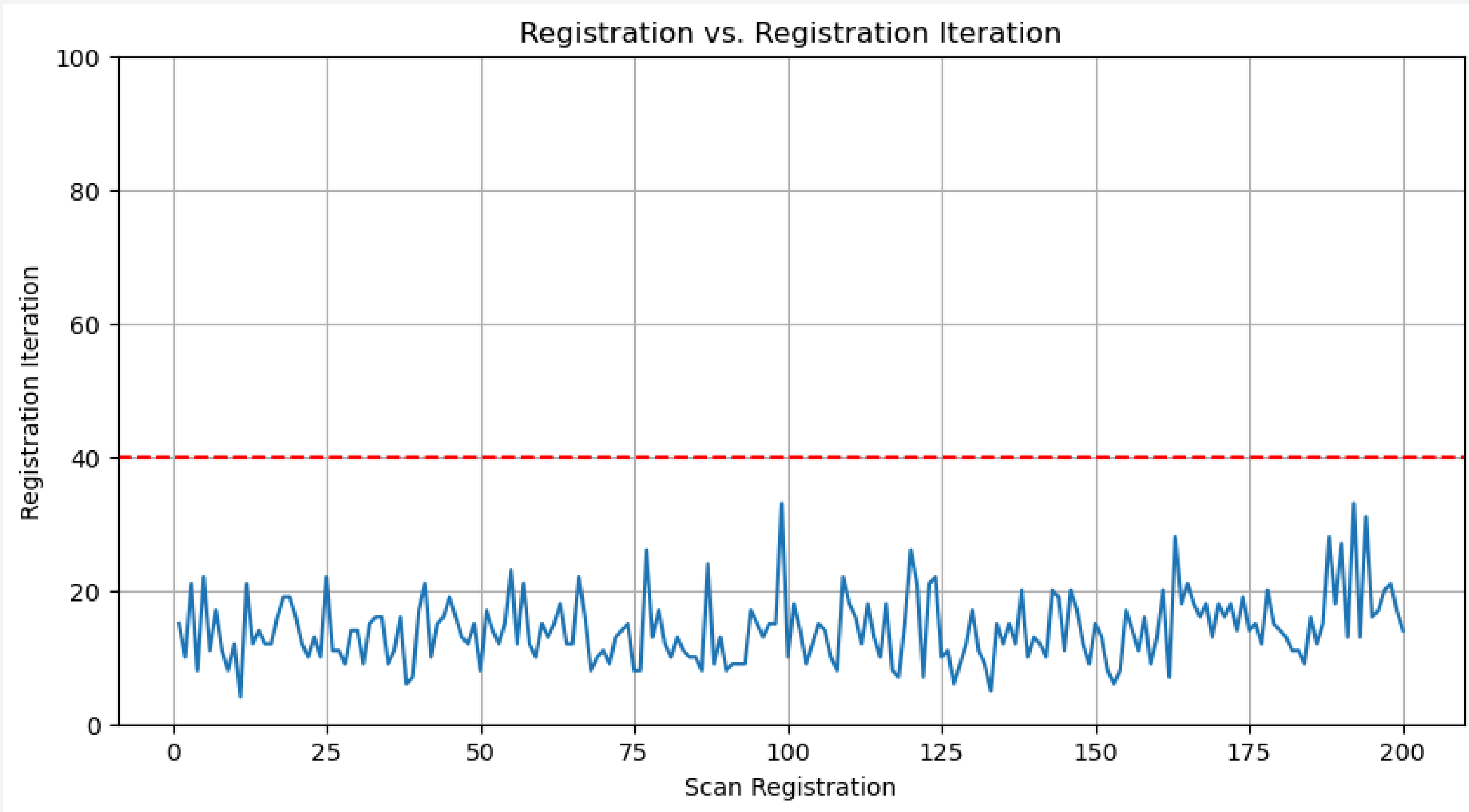
-  PCD grid
-  PCD grid loaded @ $t=t$
-  PCD grid to newly load @ $t=t$
-  PCD map grid to remove @ $t=t$

LOCAL MAP LOADING(RETRIVAL)

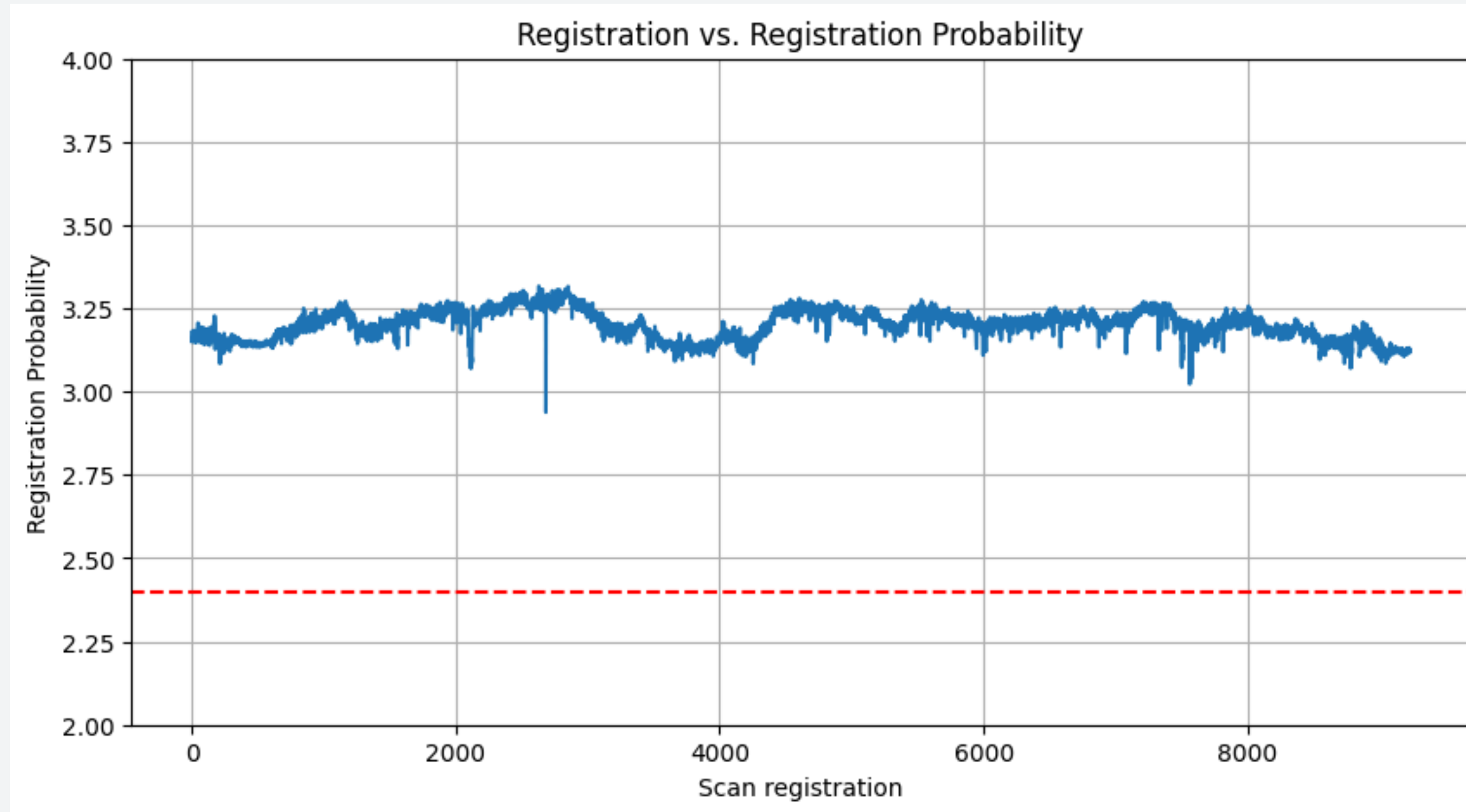
- configure local map radius considering the FOV of the lidar and some additional threshold (20m)
- Fetch grid Points only with map radius + threshold
- Next map loading after the robot moves(threshold distance)
- This is Competition Efficient - by selecting index of grid map
- Time - We are not loading everytime

SAXION DATASET 1

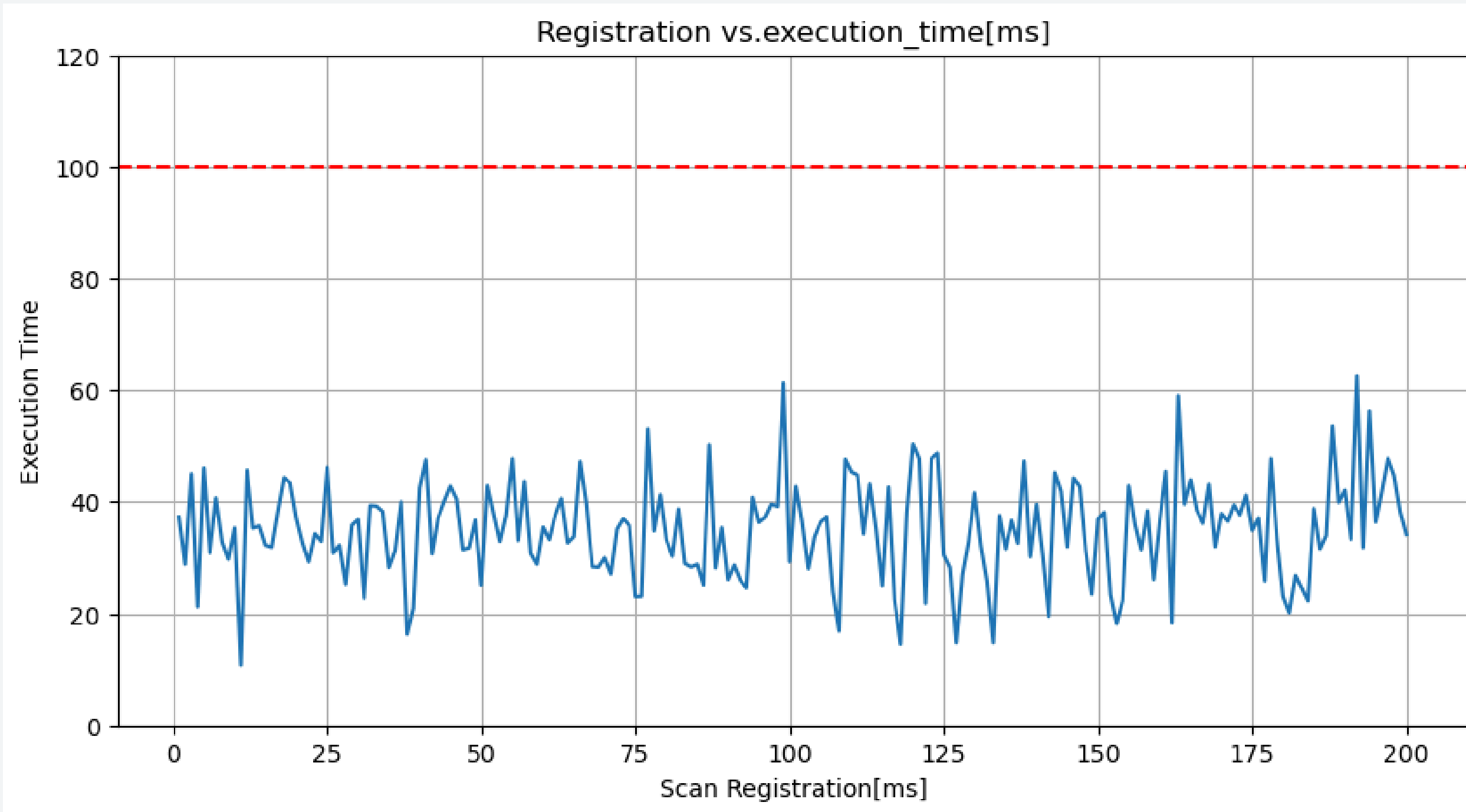
REGISTRATION ITERATION



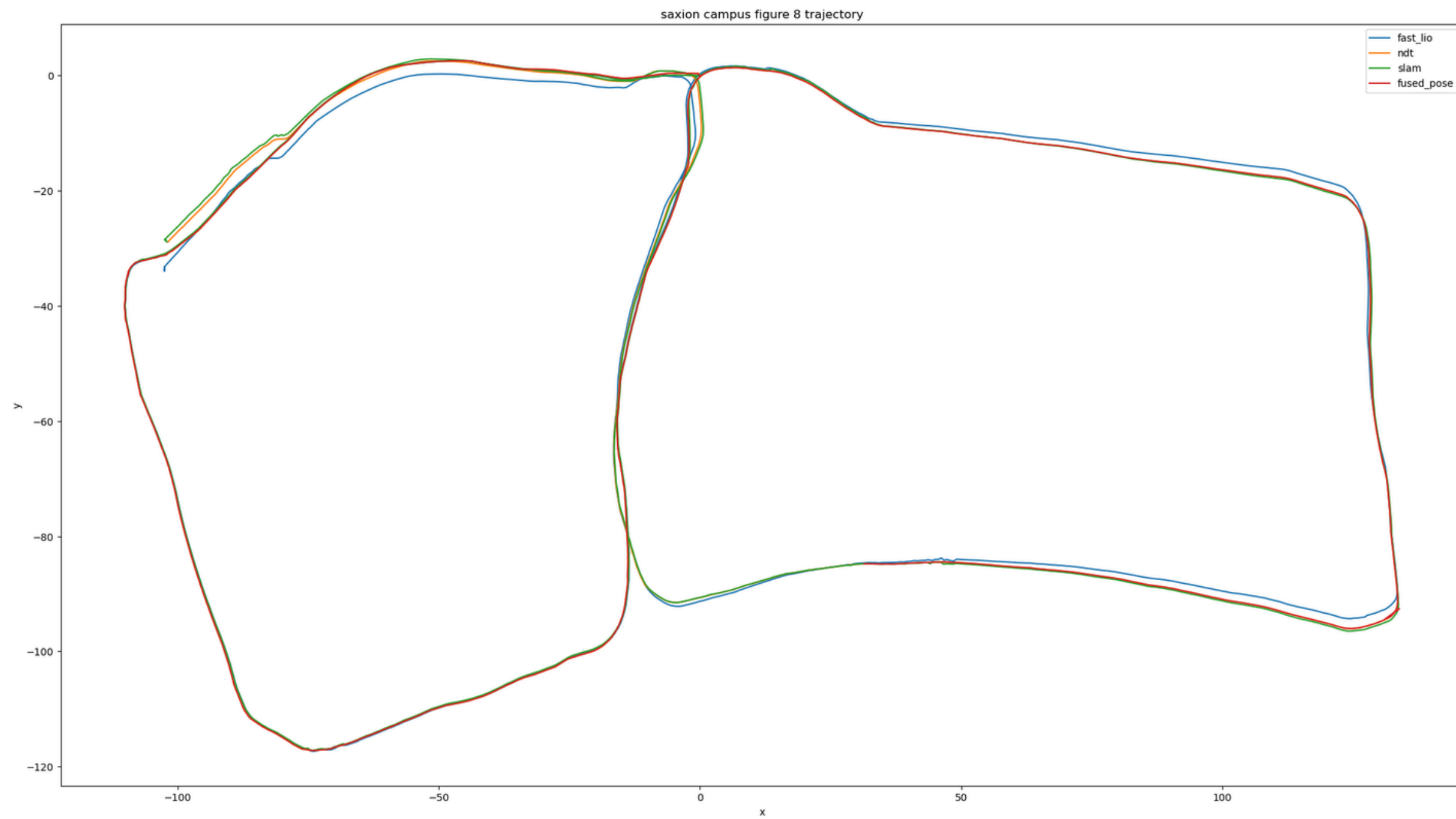
REGISTRATION SCORE

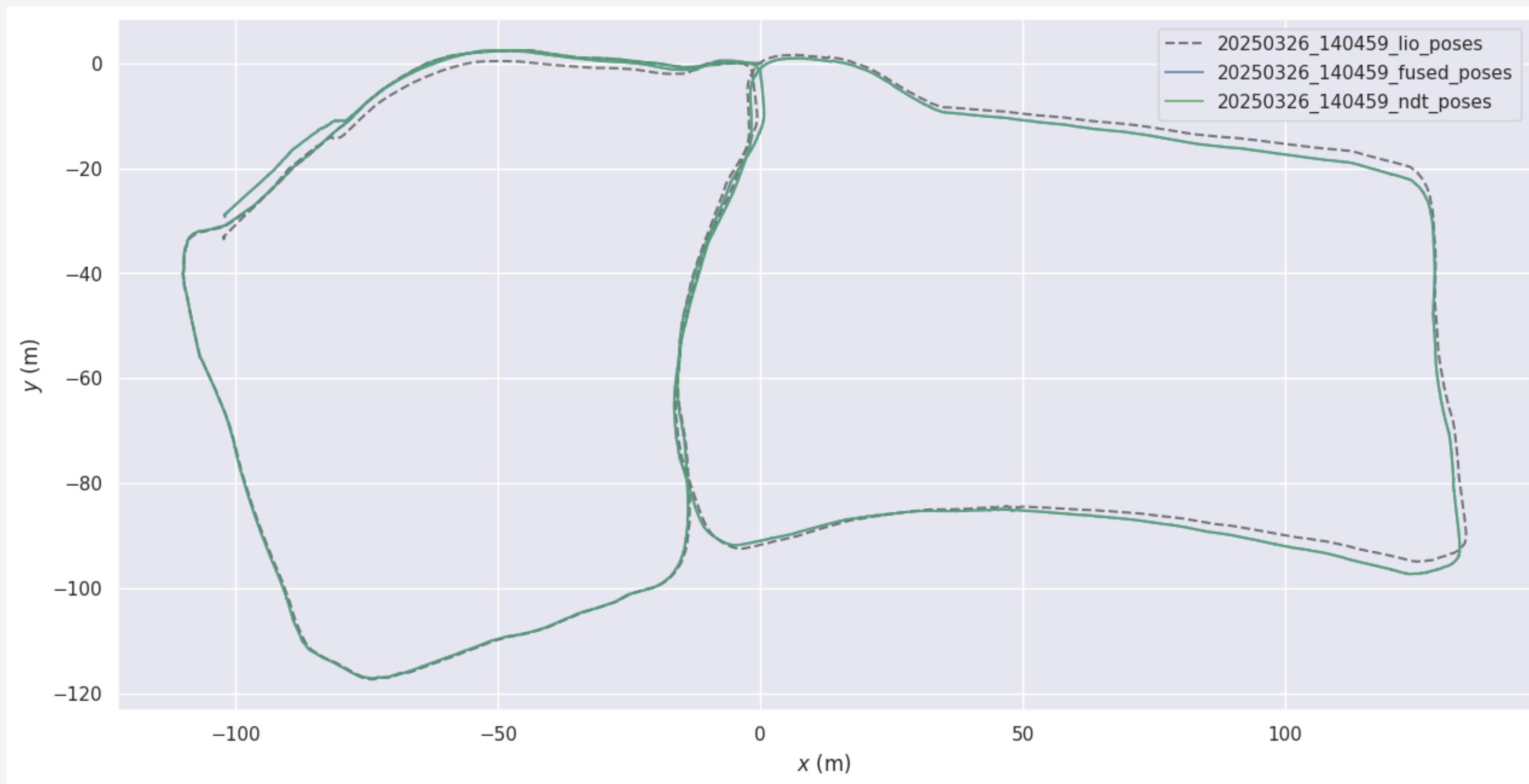


EXECUTION TIME



SAXION DATASET1_PATH



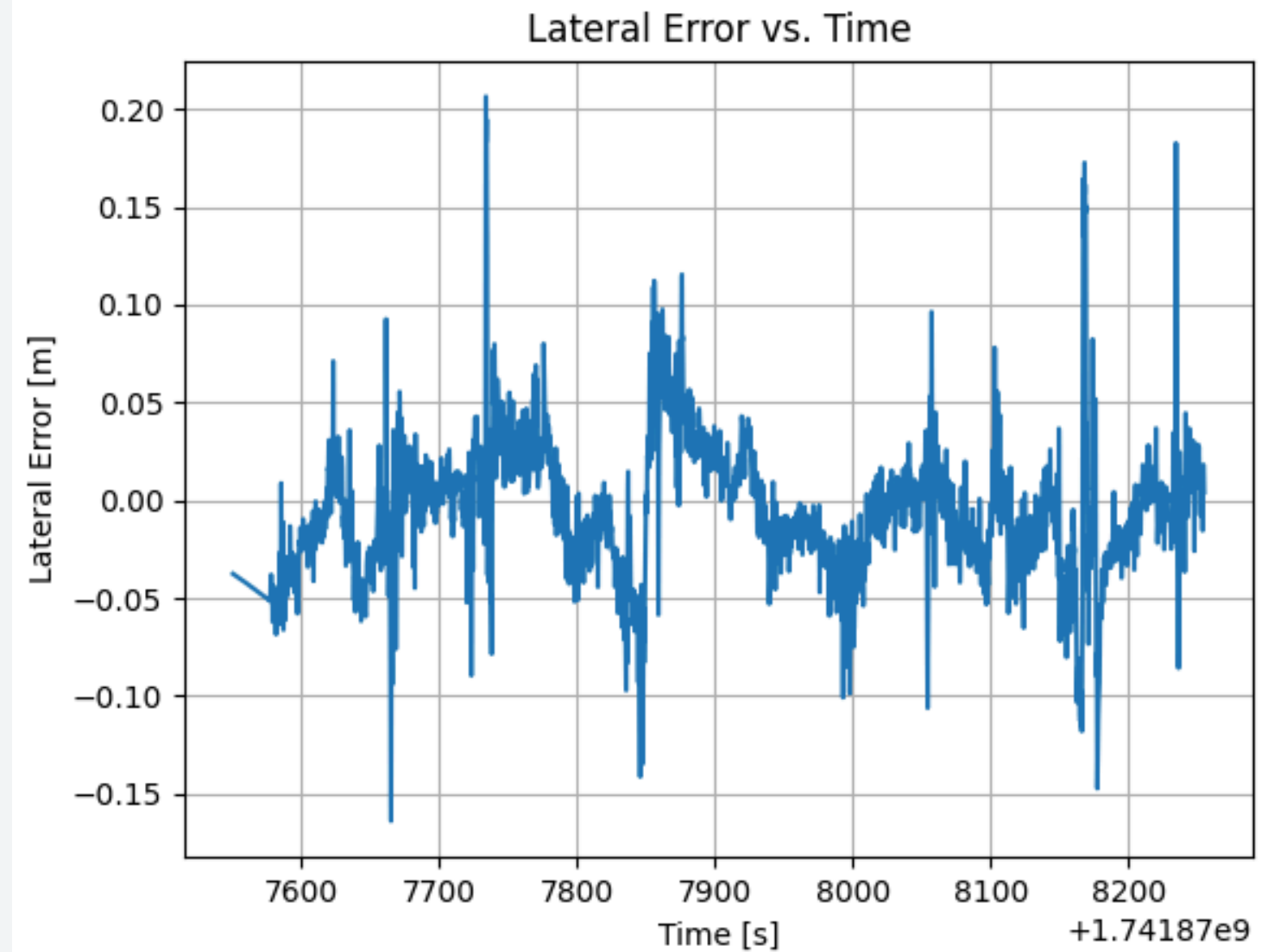
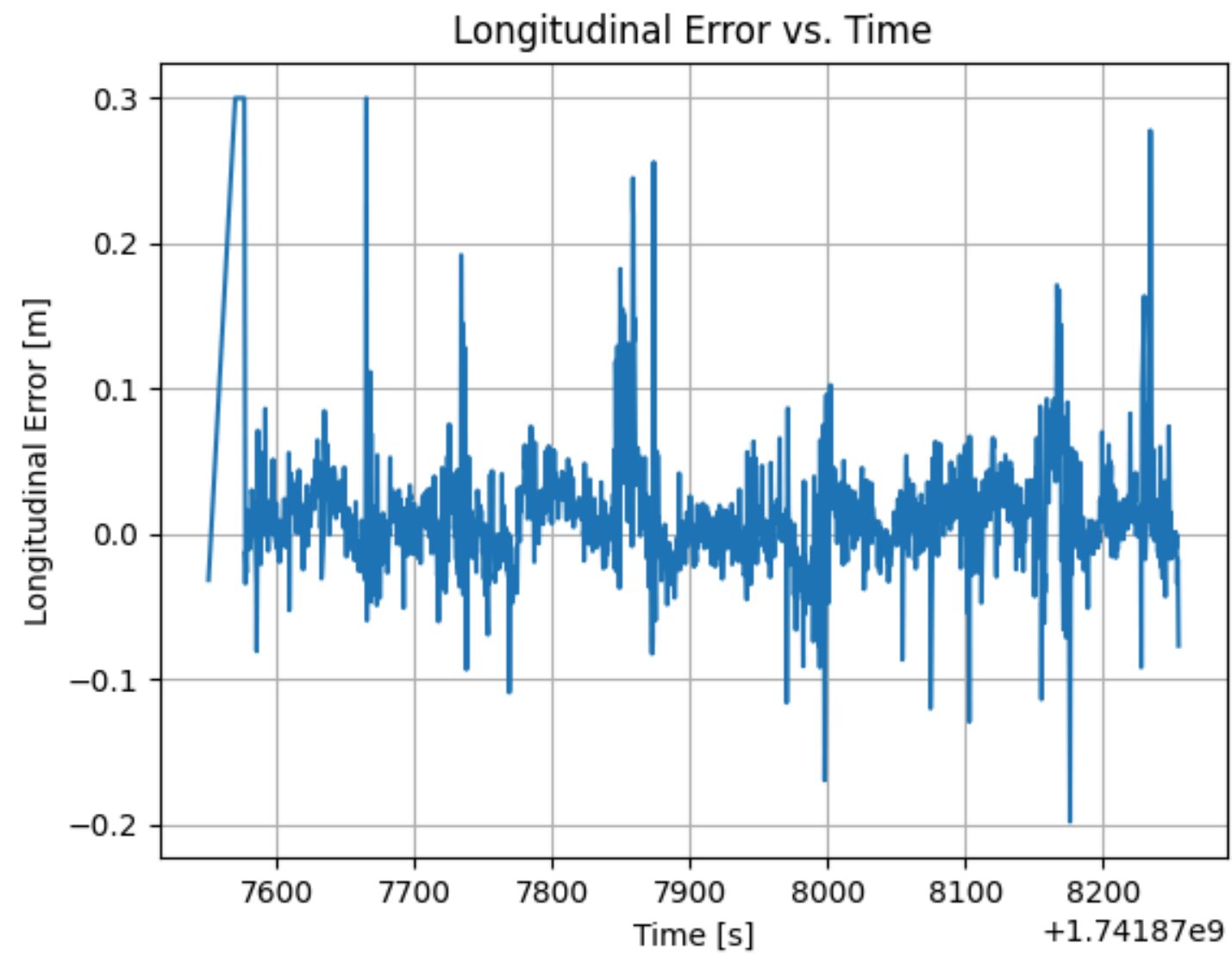


SAXION DATA 1 ERROR

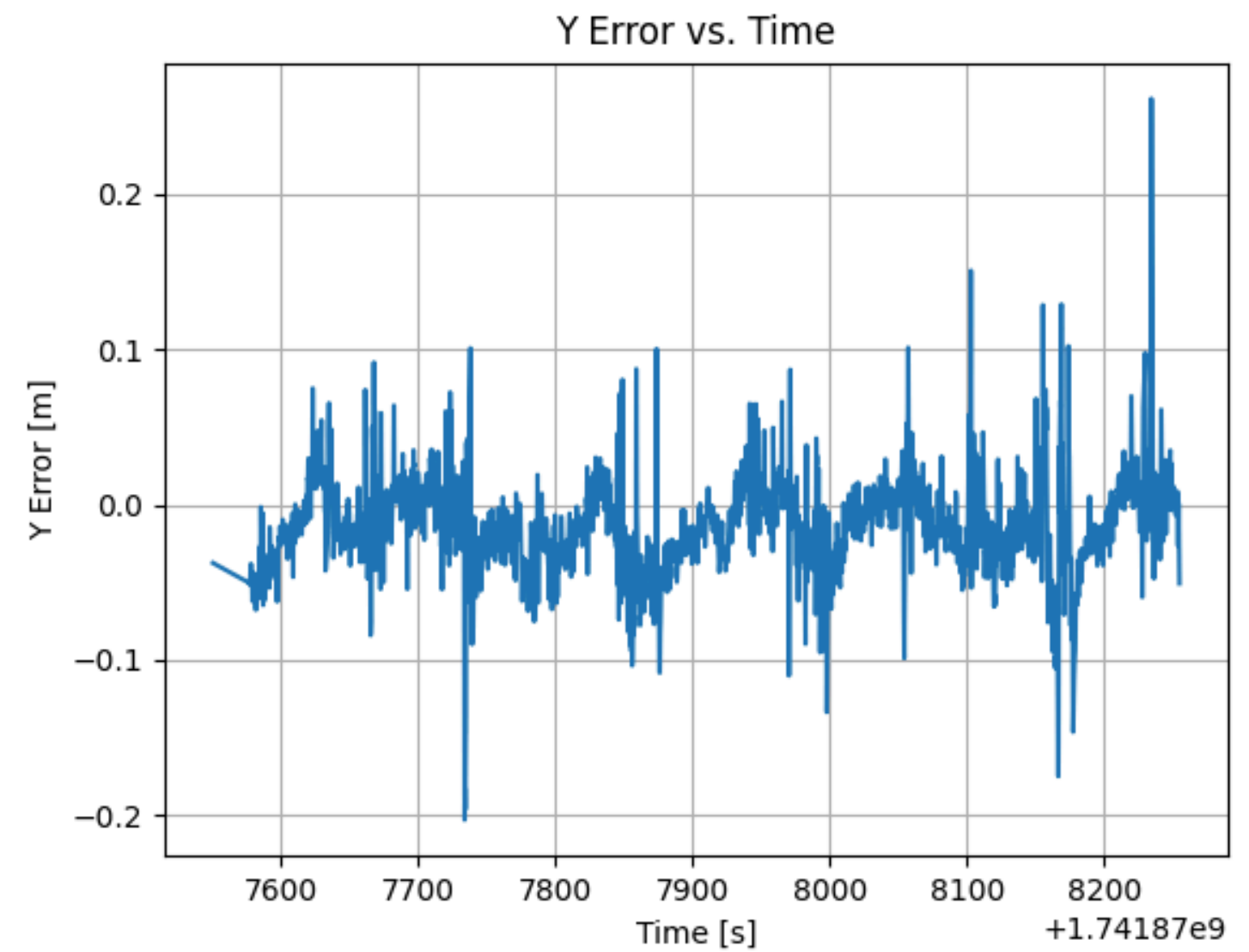
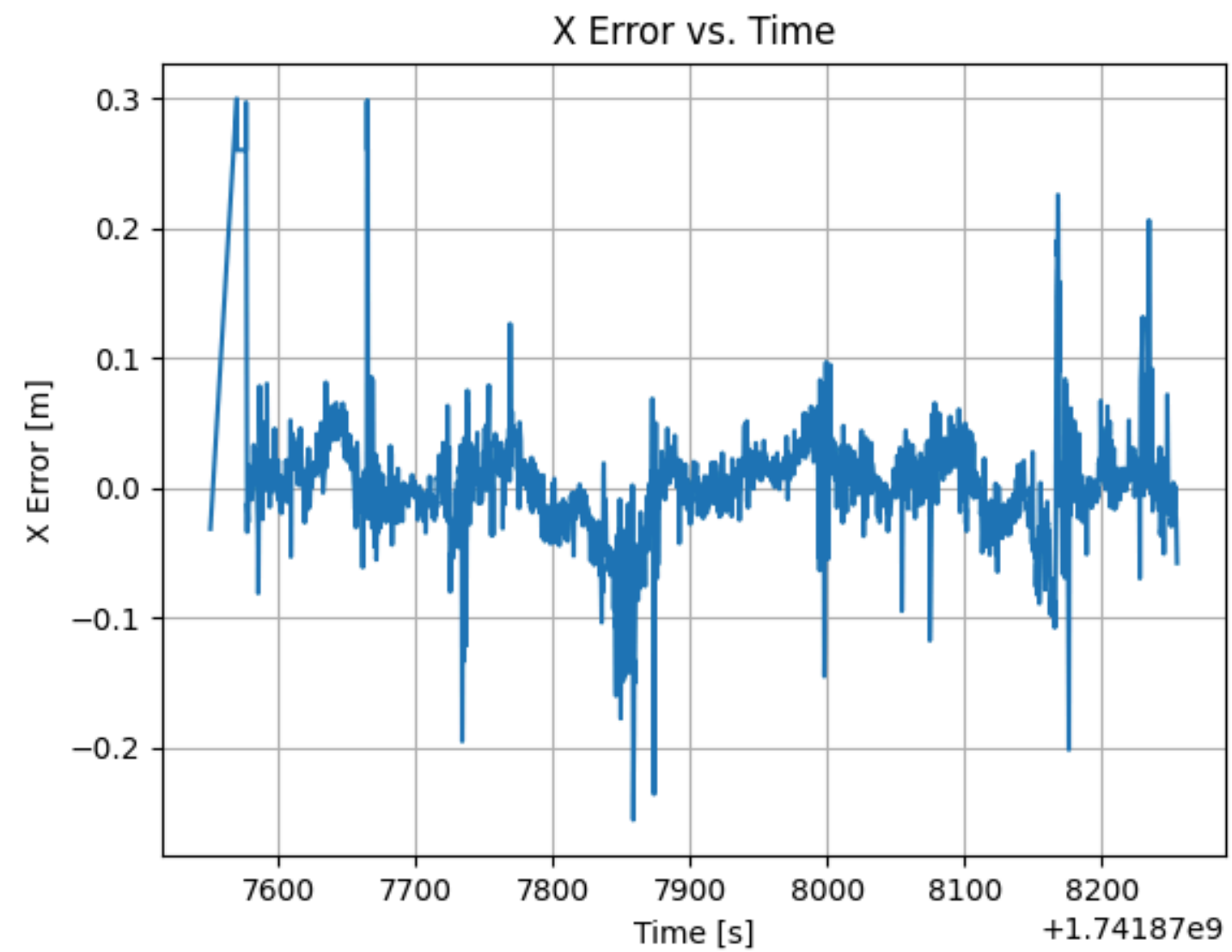
- Increase in longitudinal error

```
X Error:
  Min = 0.00000, Max = 0.29954, Mean = 0.02984, MSE = 0.00268, RMSE = 0.05176
Y Error:
  Min = 0.00000, Max = 0.26136, Mean = 0.02539, MSE = 0.00111, RMSE = 0.03339
Z Error:
  Min = 0.00000, Max = 0.47574, Mean = 0.04191, MSE = 0.00373, RMSE = 0.06107
Lateral Error:
  Min = 0.00000, Max = 0.20631, Mean = 0.02639, MSE = 0.00122, RMSE = 0.03489
Longitudinal Error:
  Min = 0.00000, Max = 0.30000, Mean = 0.02919, MSE = 0.00280, RMSE = 0.05296
elias@star: /media/elias/New Volume/autoware/data/utis$
```

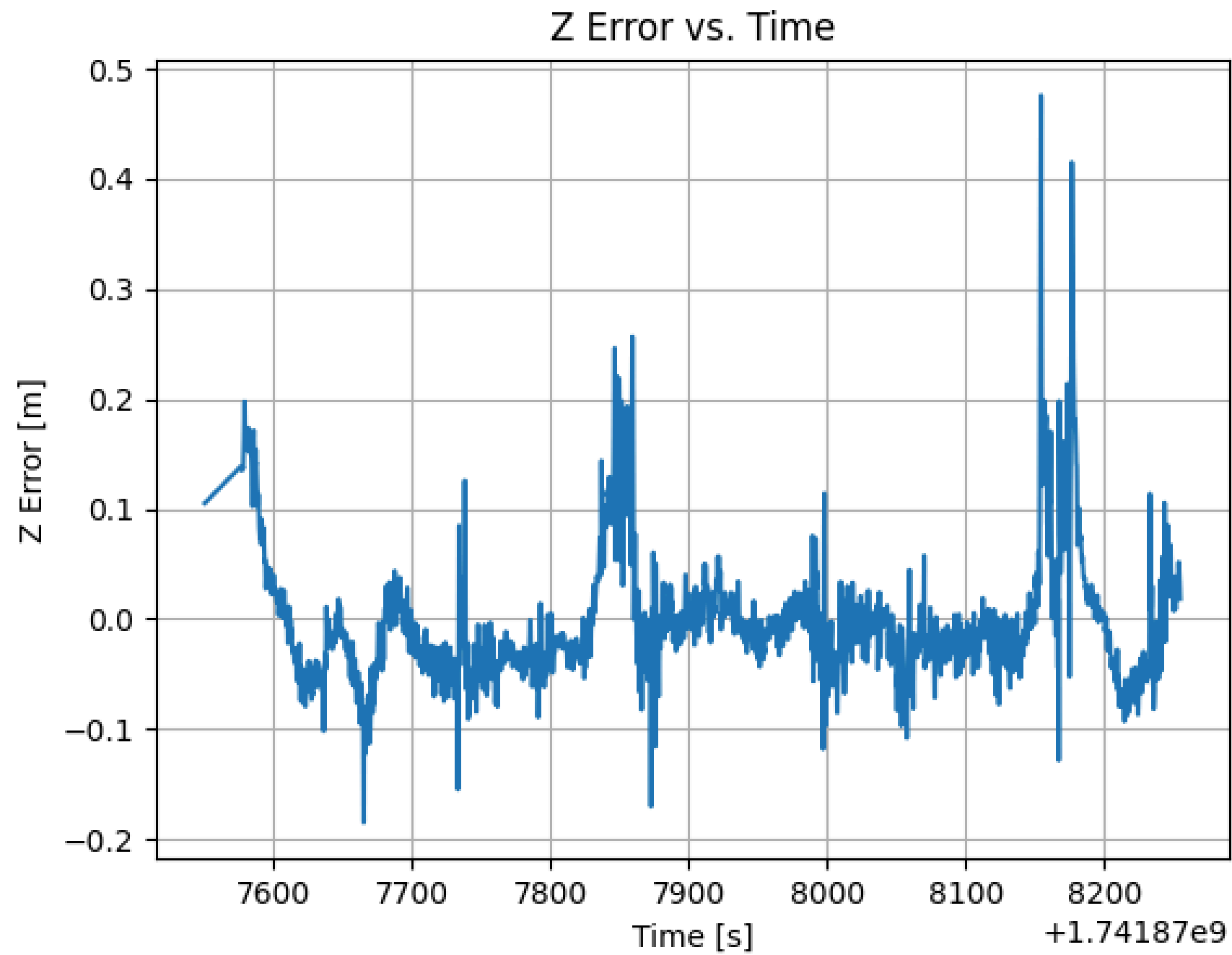
SAXION DATA 1 ERROR



SAXION DATA 1 ERROR



SAXION DATA 1 ERROR

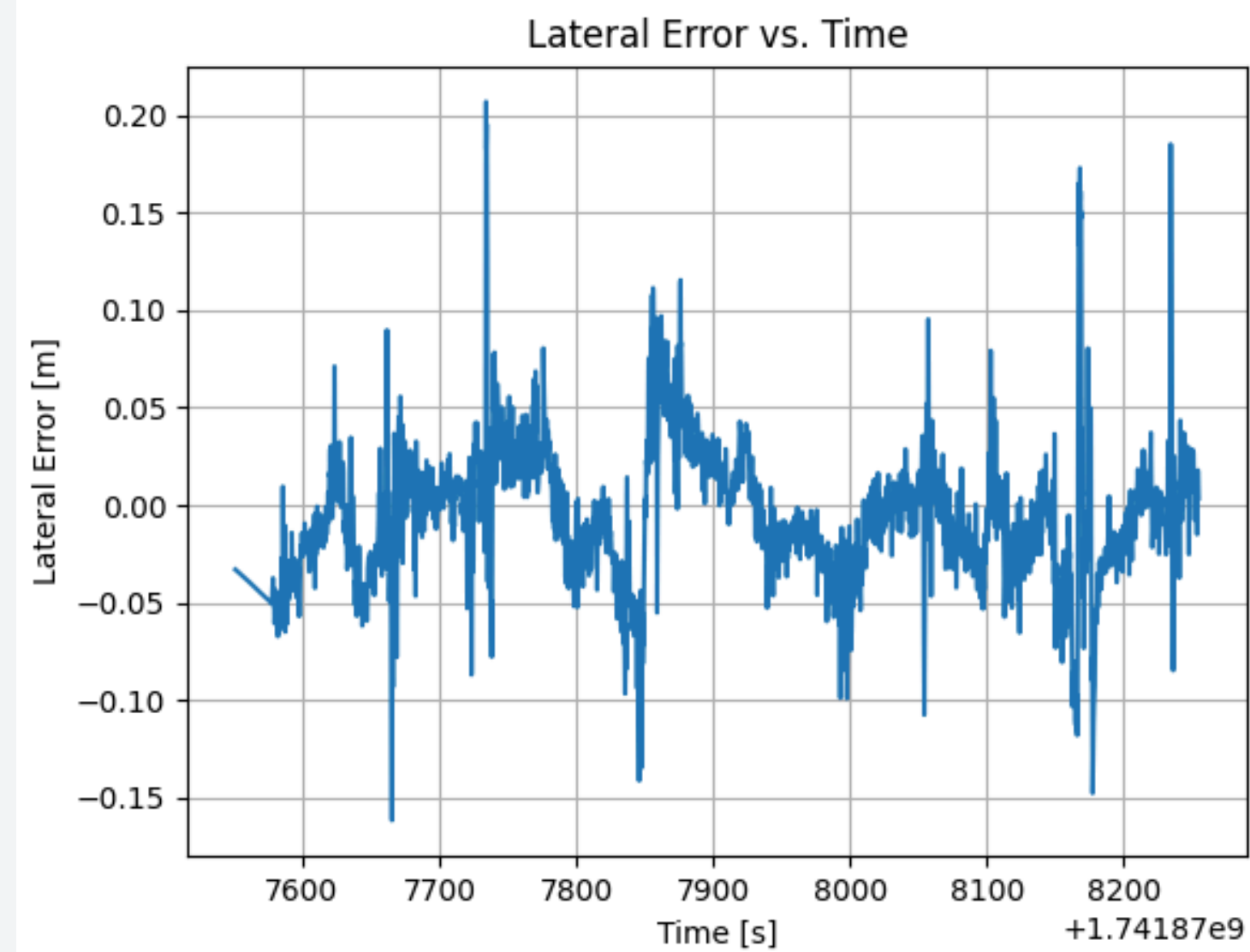
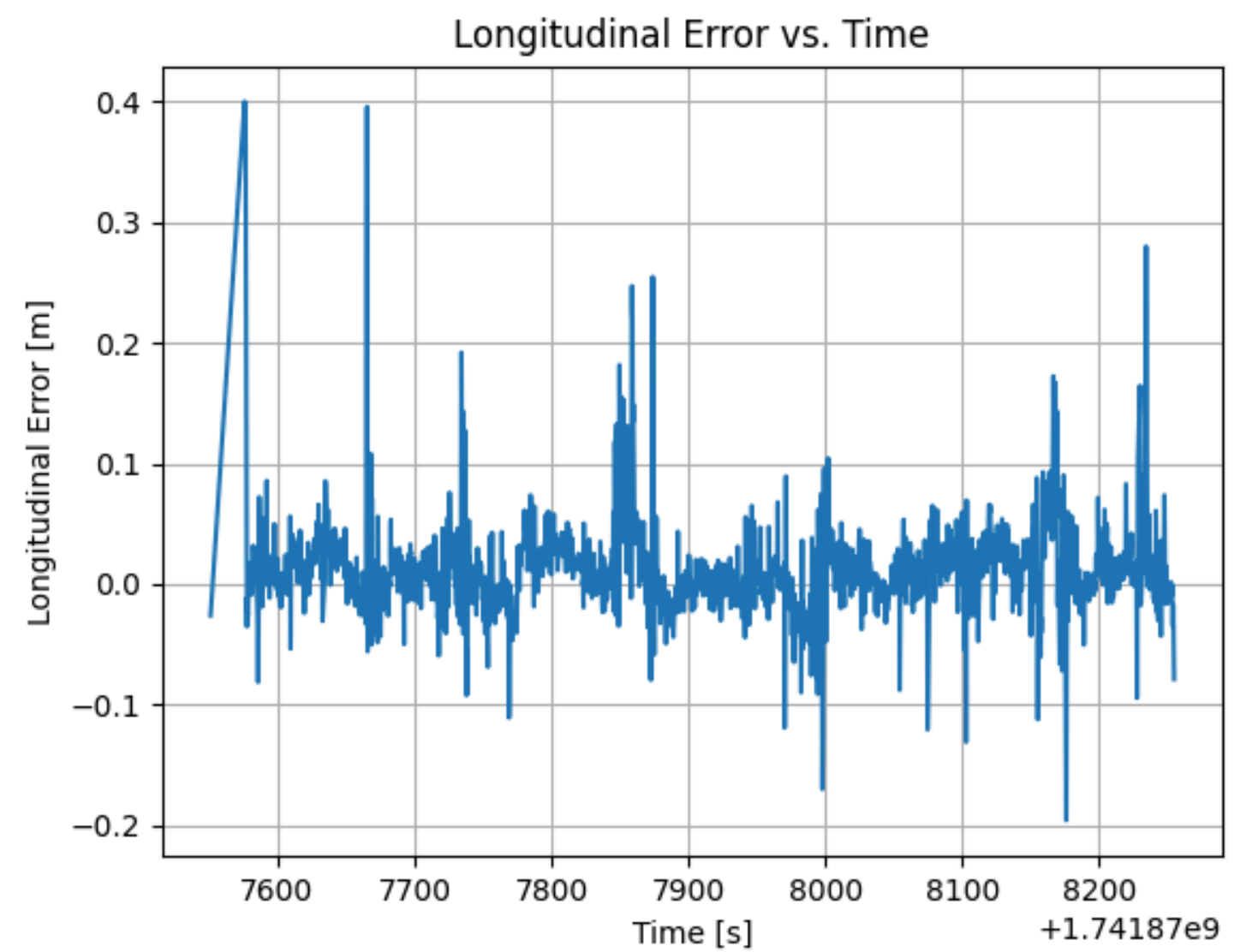


SAXION DATA 1 ERROR IN MAP 2

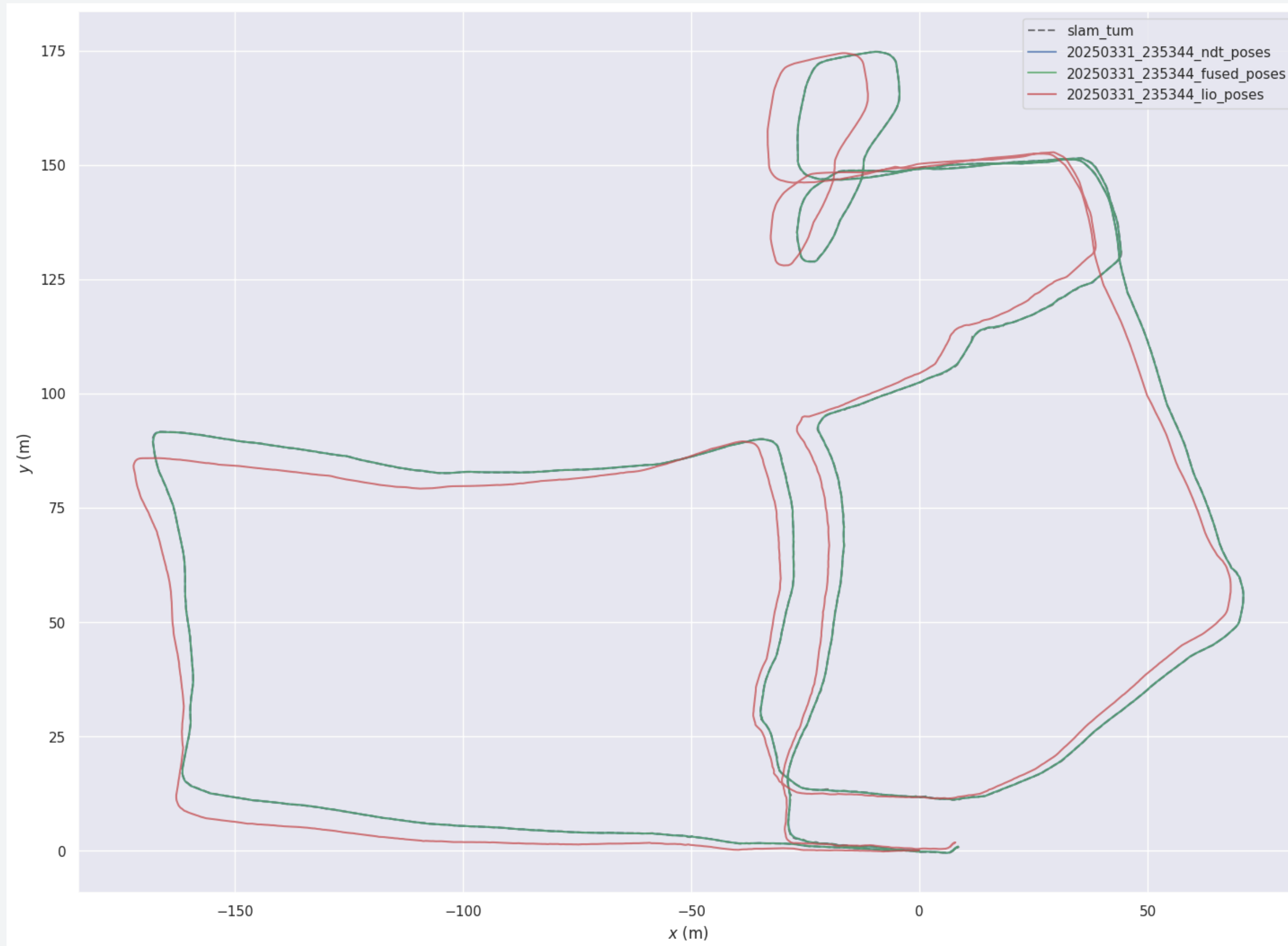
- Increase in longitudinal error

```
X Error:
  Min = 0.00000, Max = 0.39950, Mean = 0.03076, MSE = 0.00325, RMSE = 0.05701
Y Error:
  Min = 0.00000, Max = 0.26400, Mean = 0.02527, MSE = 0.00110, RMSE = 0.03324
Z Error:
  Min = 0.00000, Max = 0.47464, Mean = 0.04189, MSE = 0.00373, RMSE = 0.06107
Lateral Error:
  Min = 0.00000, Max = 0.20693, Mean = 0.02626, MSE = 0.00121, RMSE = 0.03474
Longitudinal Error:
  Min = 0.00000, Max = 0.39991, Mean = 0.02973, MSE = 0.00315, RMSE = 0.05613
```





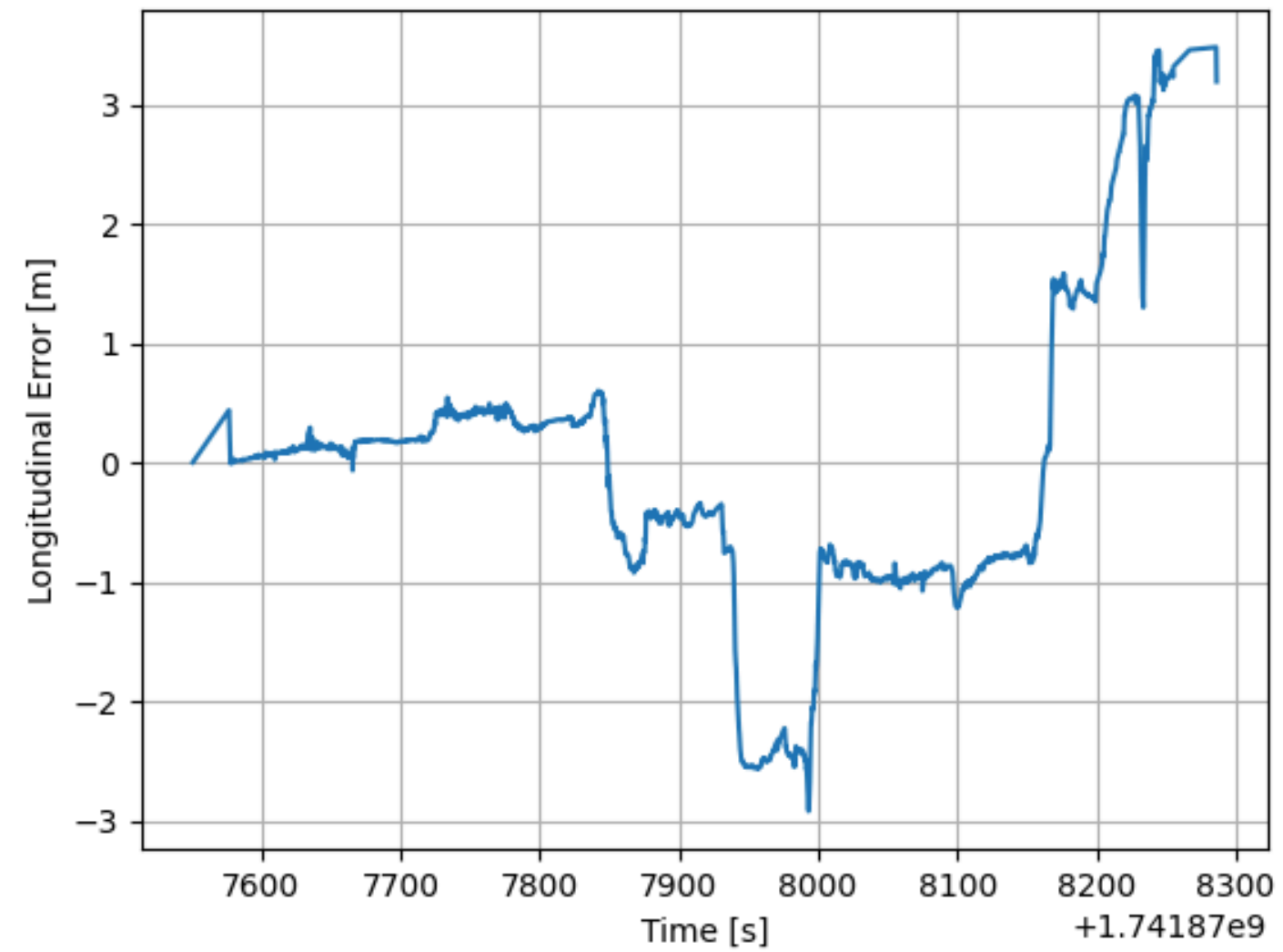
SAXION2 PATH ERROR



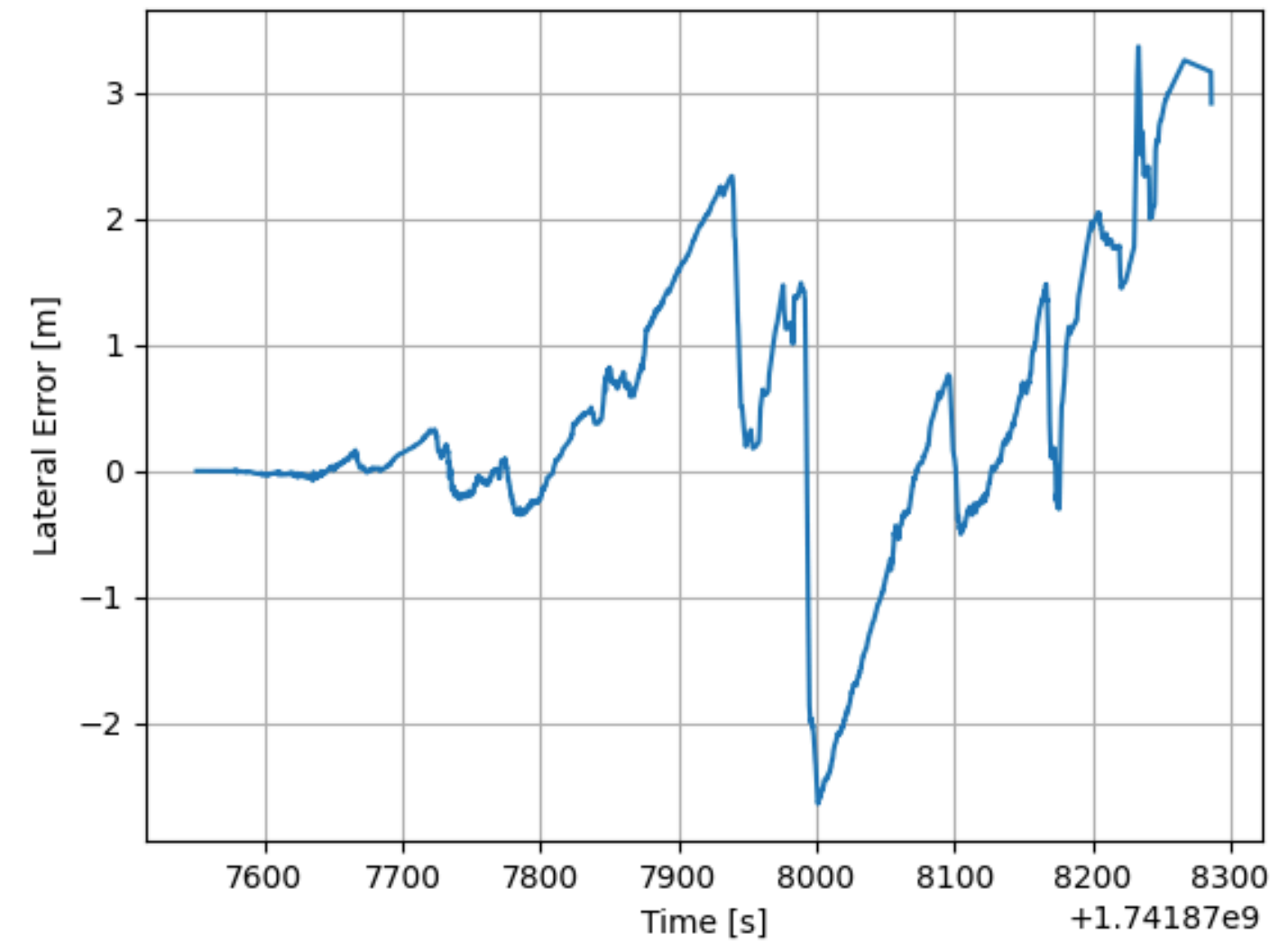
SAXION2 PATH LIO ERROR

```
elias@star:/media/elias/New Volume/autoware/data/utis$ python3 interpolate_both.py ../Finall_results_and_log/
xion_fig_8_log/31b/20250331_224846_lio_poses ../Finall_results_and_log/saxion_fig_8_log/evo/slam_tum.txt
X Error:
  Min = 0.00002, Max = 1.58477, Mean = 0.51703, MSE = 0.47117, RMSE = 0.68642
Y Error:
  Min = 0.00000, Max = 4.75347, Mean = 1.14042, MSE = 2.96270, RMSE = 1.72125
Z Error:
  Min = 0.00006, Max = 8.99249, Mean = 4.08625, MSE = 25.58540, RMSE = 5.05820
Lateral Error:
  Min = 0.00000, Max = 3.36780, Mean = 0.84498, MSE = 1.55867, RMSE = 1.24847
Longitudinal Error:
  Min = 0.00006, Max = 3.48164, Mean = 0.96613, MSE = 1.87519, RMSE = 1.36938
```

Longitudinal Error vs. Time



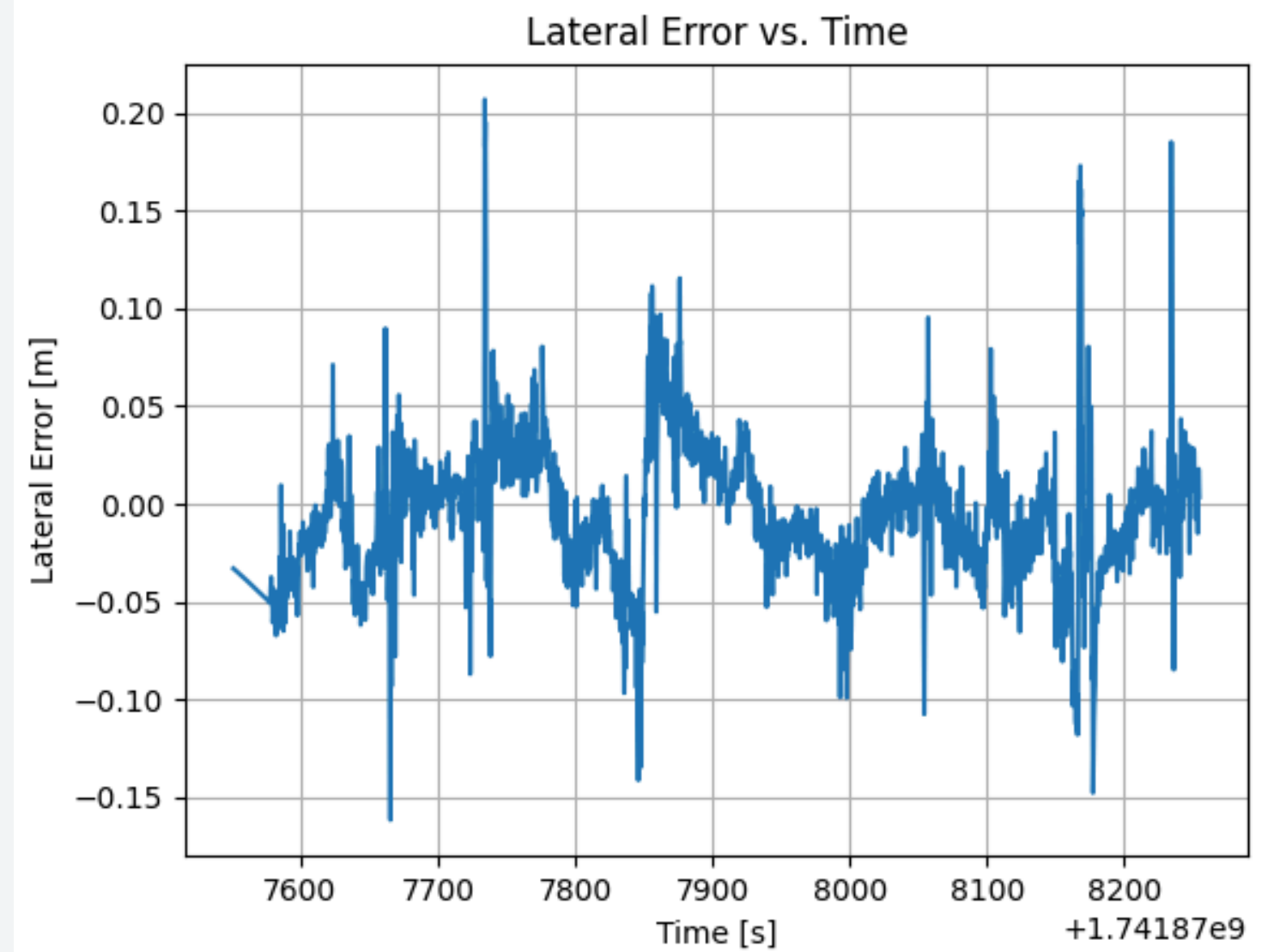
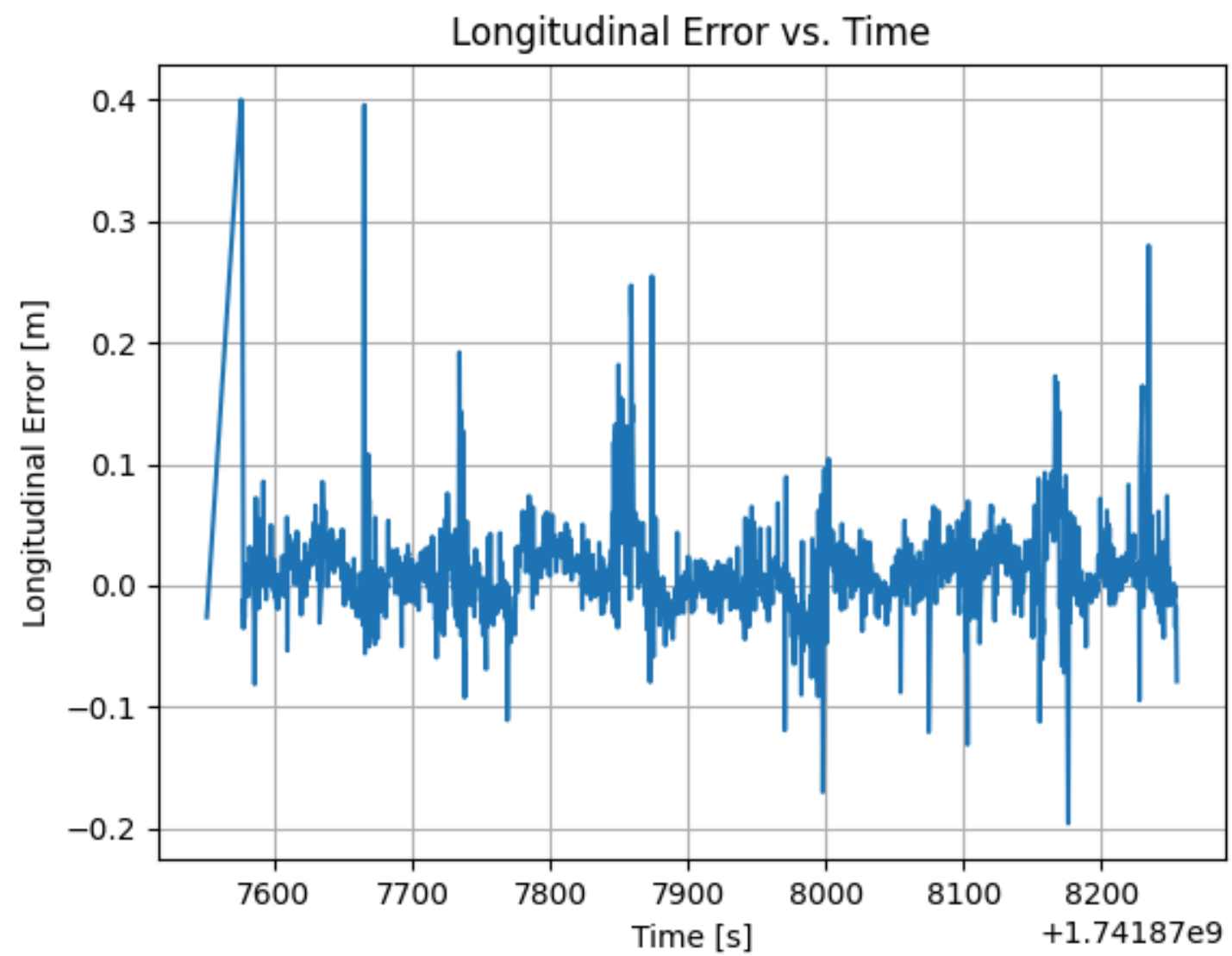
Lateral Error vs. Time



SAXION2 PATH ERROR

```
elias@star:/media/elias/New Volume/autoware/data/utis$ python3 interpolate_both_copy.py .  
og/saxion_super/31/20250331_235344_ndt_poses ../Finall_results_and_log/saxion_super/slam_t  
X Error:  
  Min = 0.00001, Max = 0.32655, Mean = 0.06738, MSE = 0.00725, RMSE = 0.08517  
Y Error:  
  Min = 0.00002, Max = 0.34353, Mean = 0.05829, MSE = 0.00584, RMSE = 0.07645  
Z Error:  
  Min = 0.00000, Max = 0.37878, Mean = 0.06000, MSE = 0.00729, RMSE = 0.08541  
Lateral Error:  
  Min = 0.00000, Max = 0.27409, Mean = 0.03271, MSE = 0.00197, RMSE = 0.04435  
Longitudinal Error:  
  Min = 0.00000, Max = 0.31963, Mean = 0.09021, MSE = 0.01102, RMSE = 0.10499
```

SAXION2 PATH ERROR



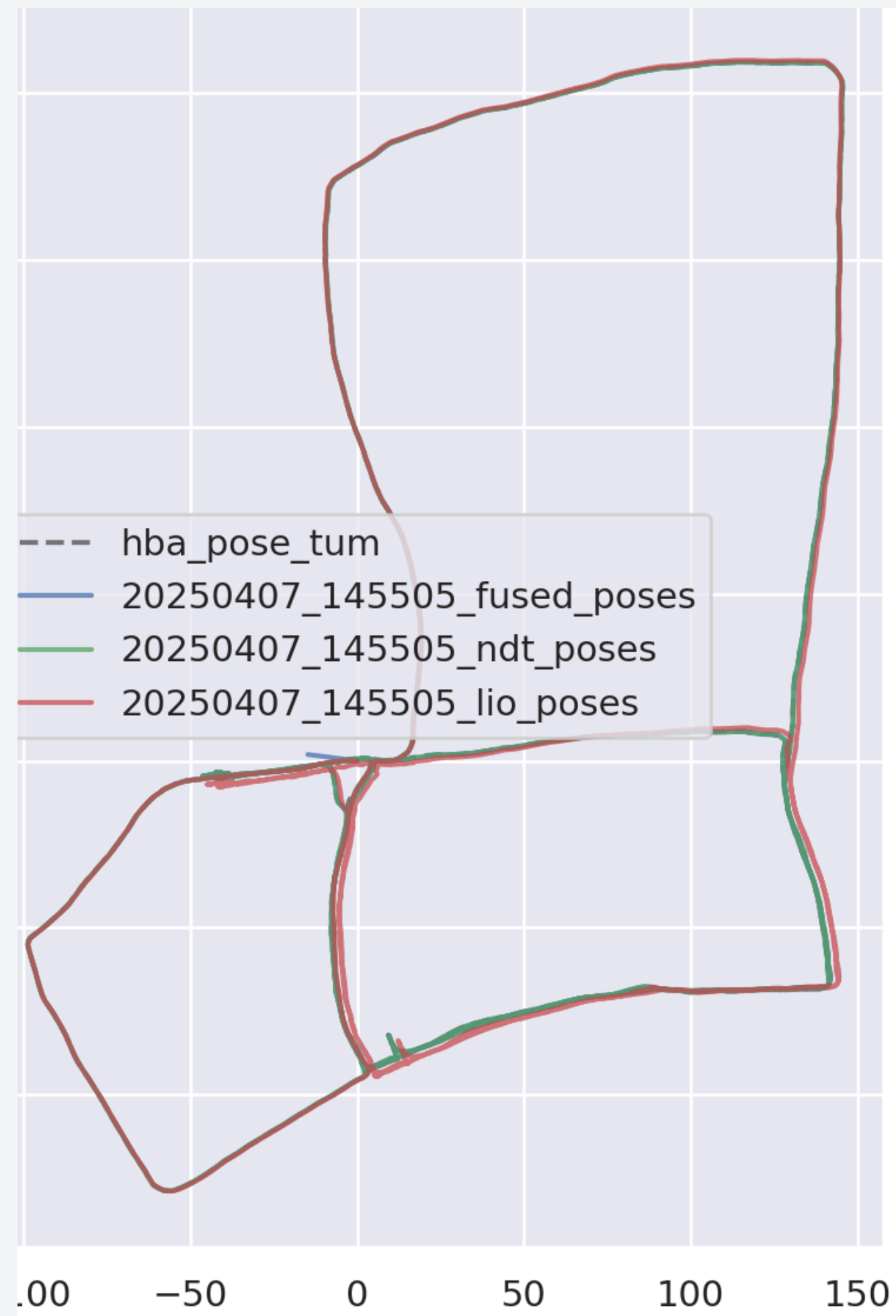
PROGRESS

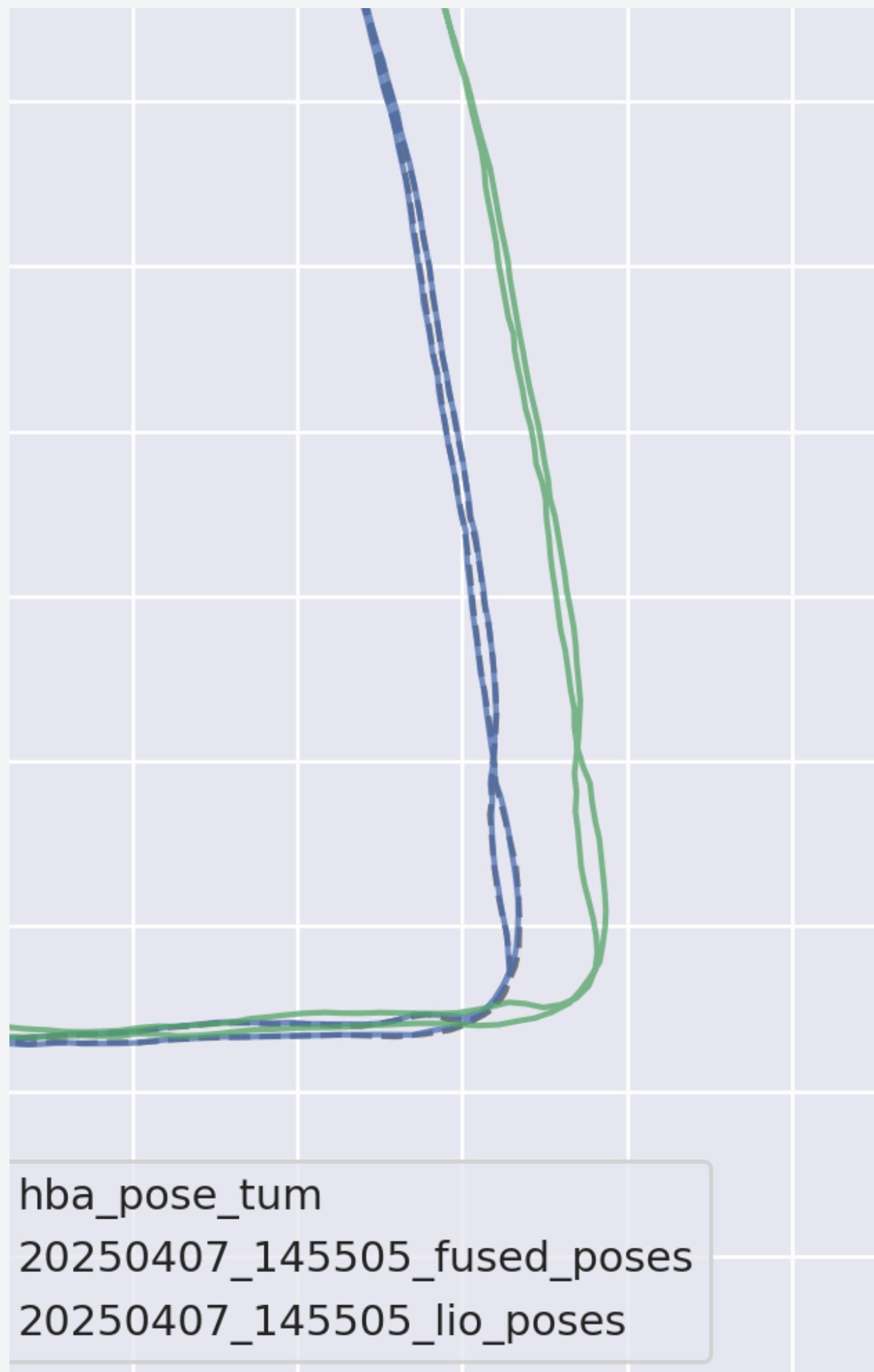
LONG TRAJECTORY

ADD NOISE(SCENARIO)

TRAJECTORY USING DIFFERENT MAP

LONG TRAJECTORY

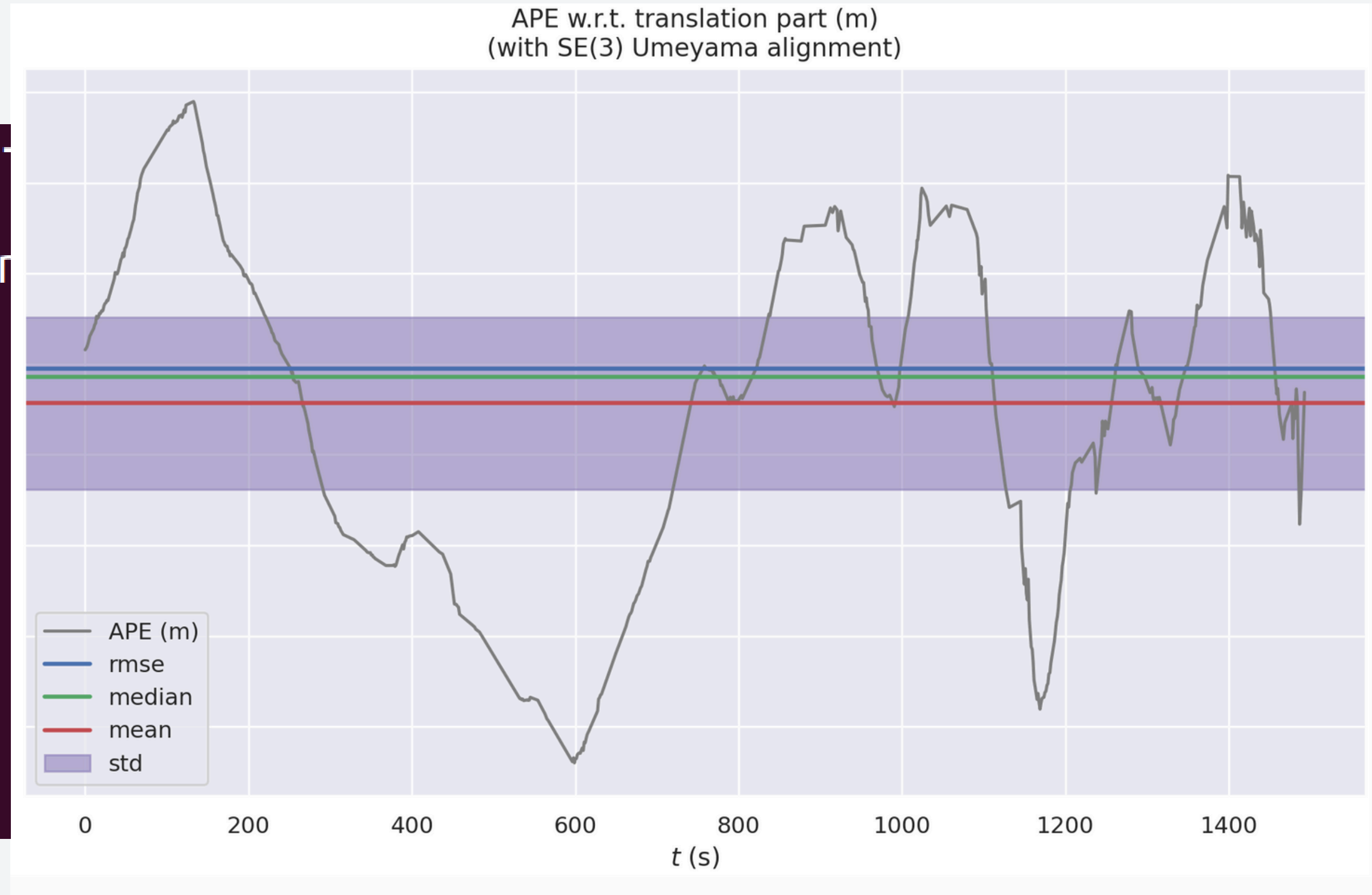




ATE LIO ERROR

E w.r.t. translation part
th SE(3) Umeyama alignmen

max	1.973925
mean	1.142012
median	1.213828
min	0.150460
rmse	1.236673
sse	747.857246
std	0.474520

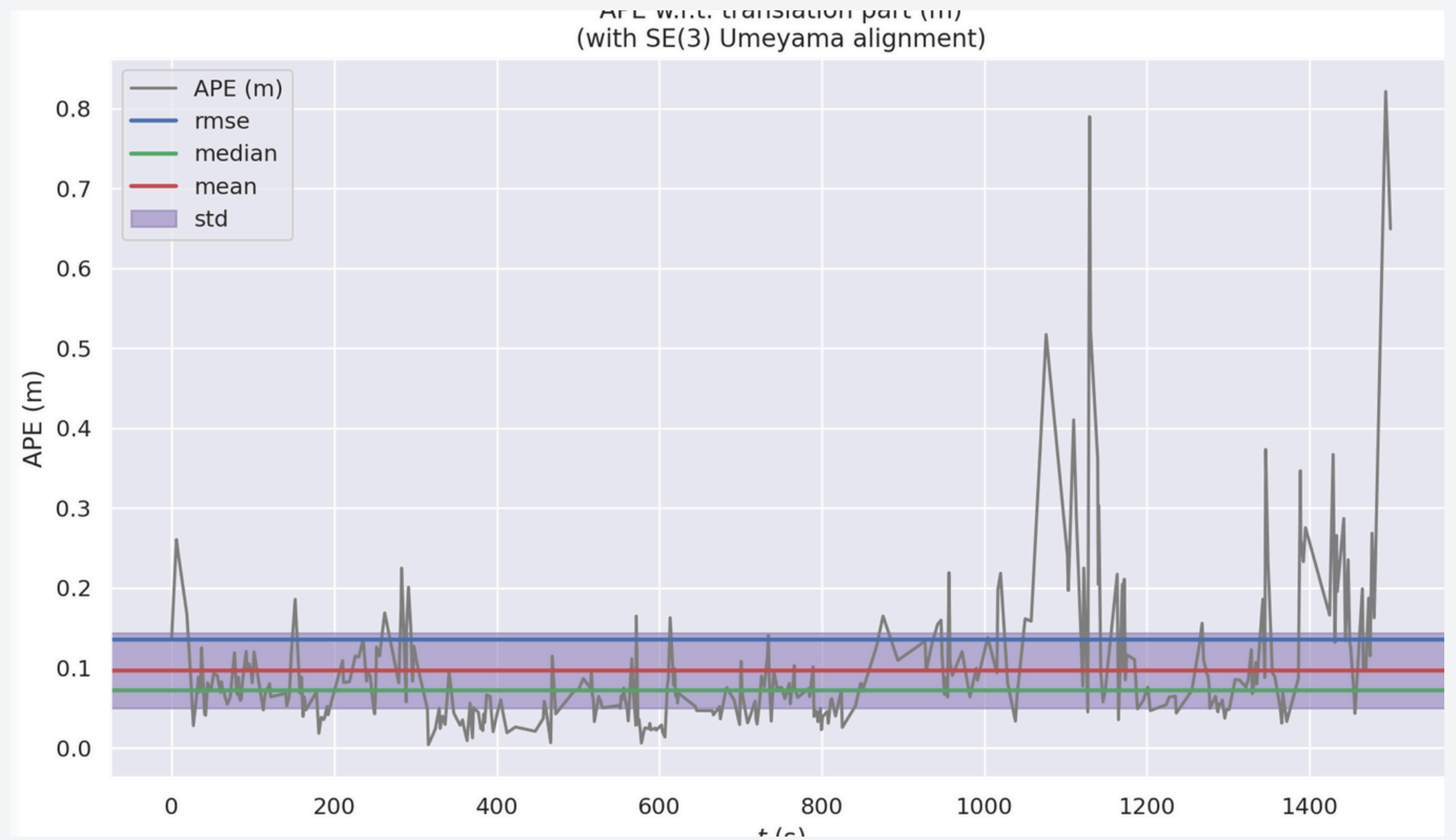


ATE FUSED ERROR

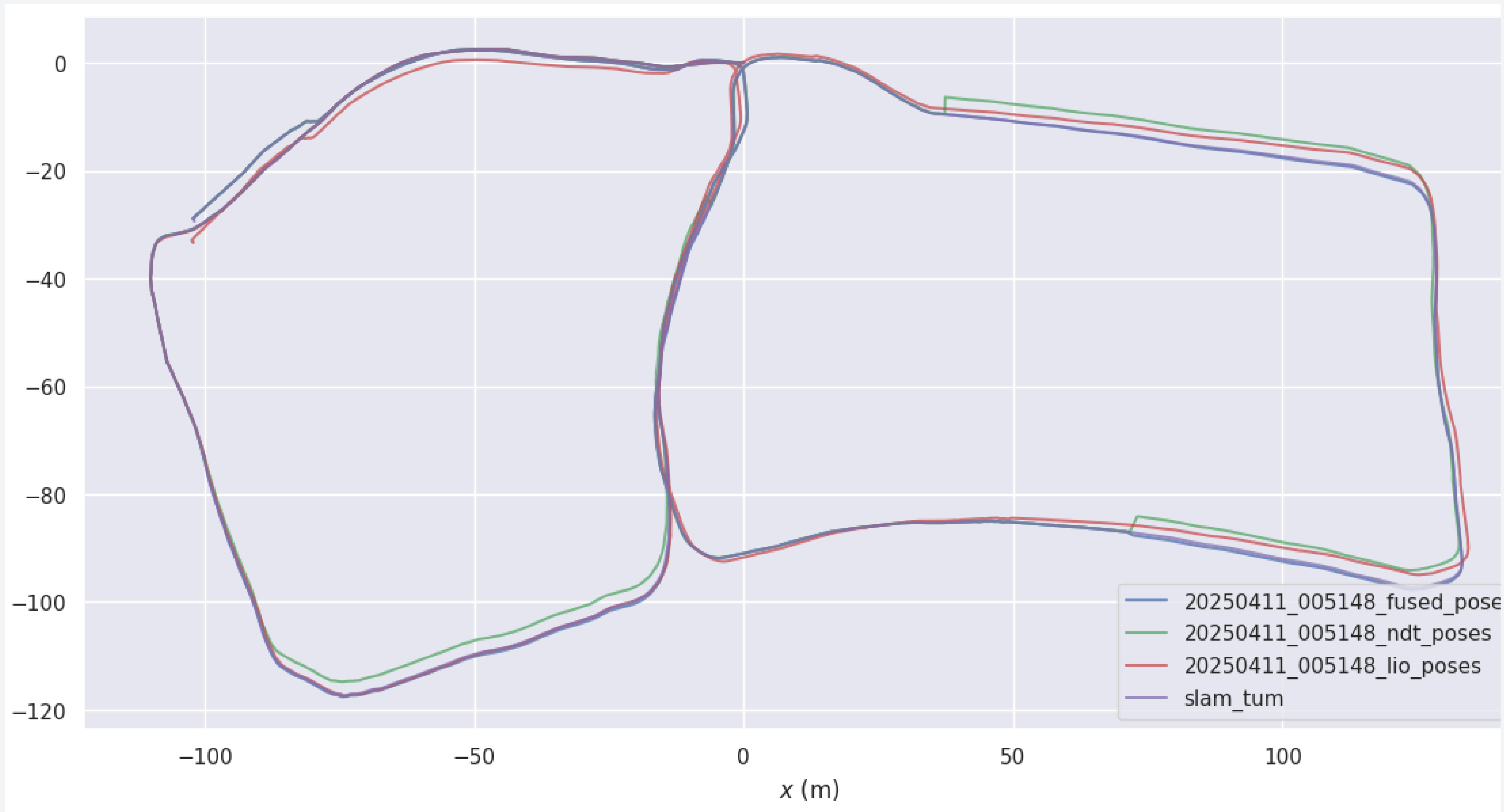
```
-----  
r.t. translation part (m)  
SE(3) Umeyama alignment)
```

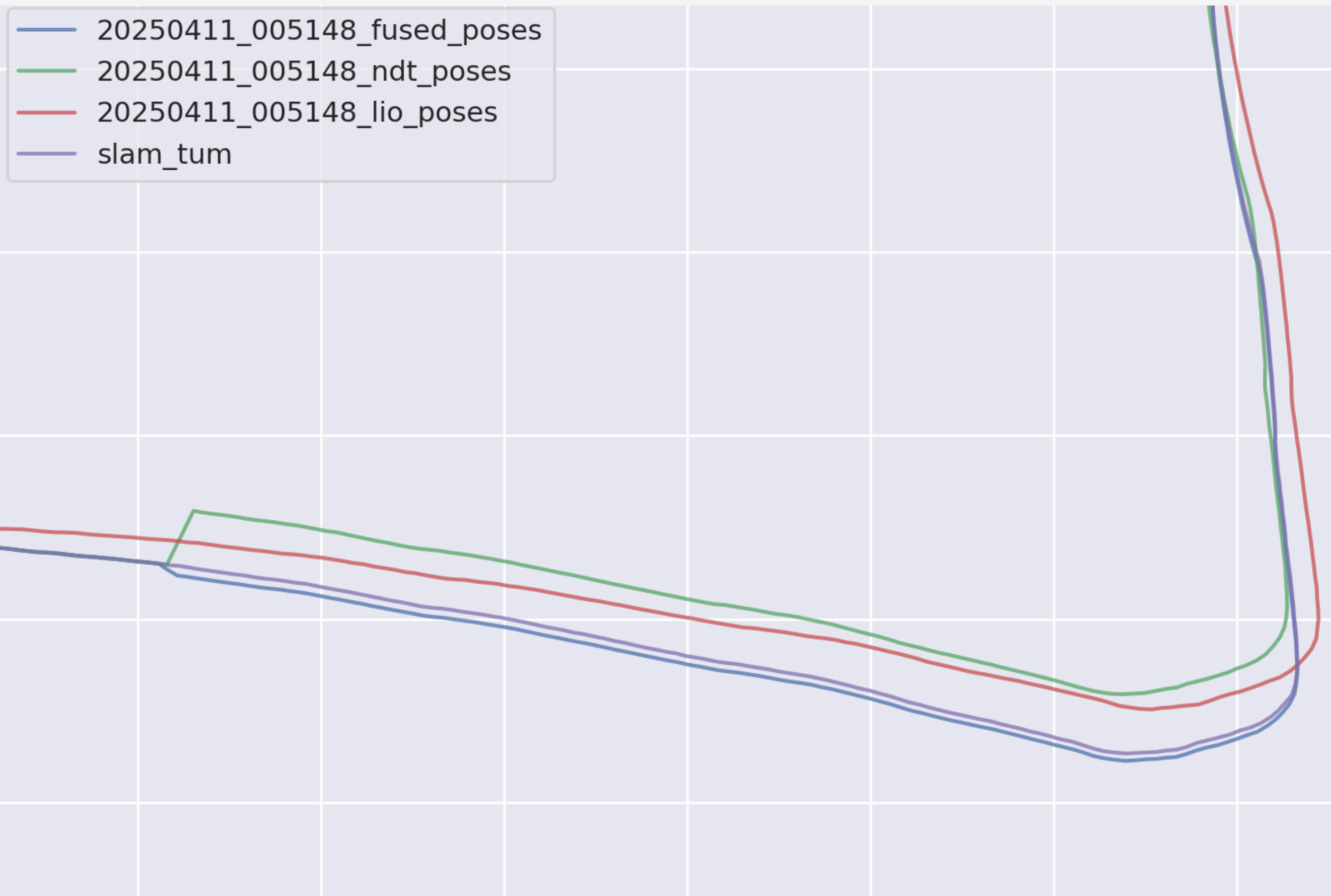
max	0.821155
mean	0.096915
median	0.071633
min	0.003928
rmse	0.135116
sse	6.663531
std	0.094147

```
-----  
ng results...
```



ADD NOISE TO MATCHING





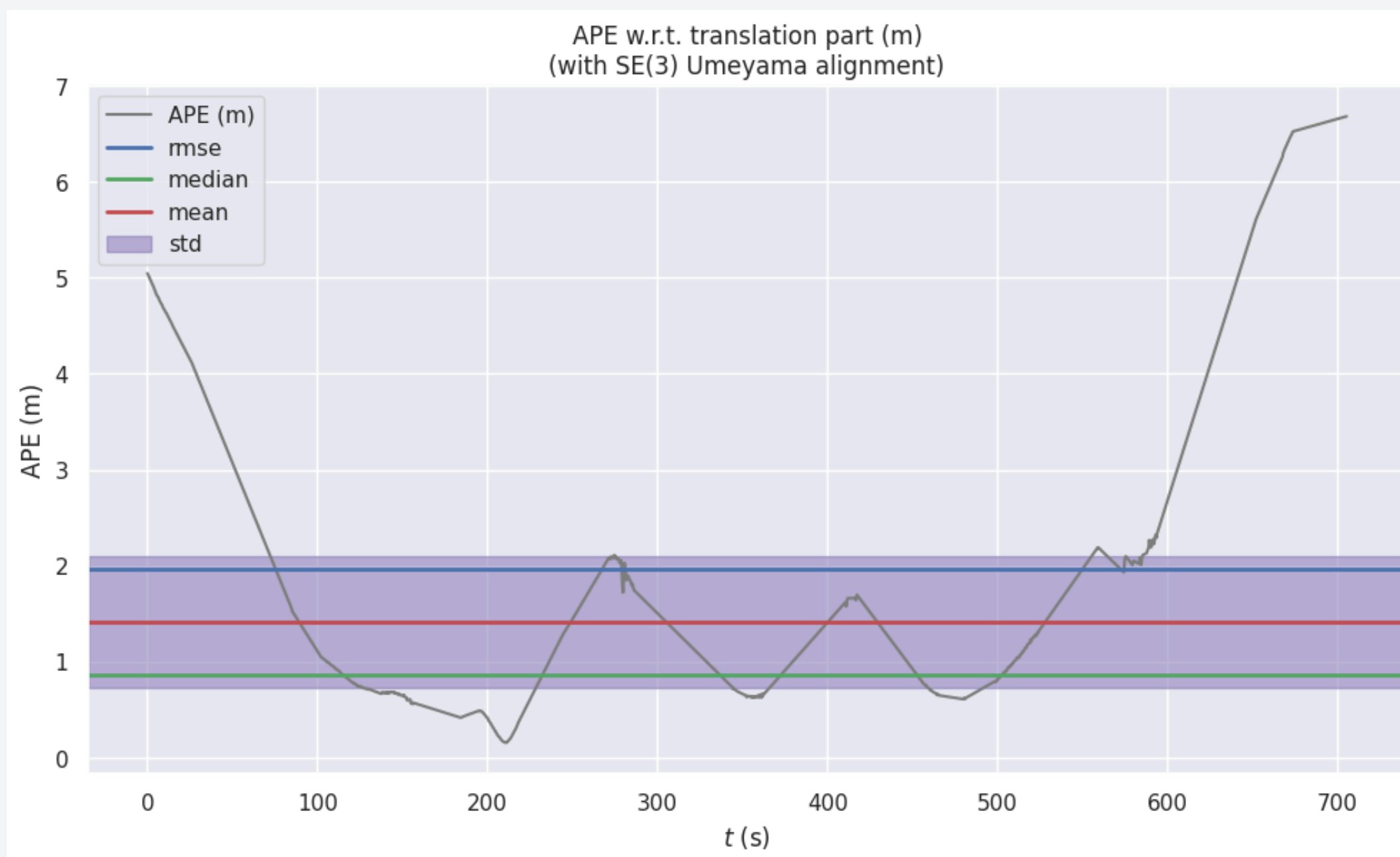
LIO

Compared 473 absolute pose pairs.
Calculating APE for translation part

APE w.r.t. translation part (m)
(with SE(3) Umeyama alignment)

max	6.681470
mean	1.413903
median	0.861592
min	0.157450
rmse	1.970284
sse	1836.195119
std	1.372187

Plotting results...



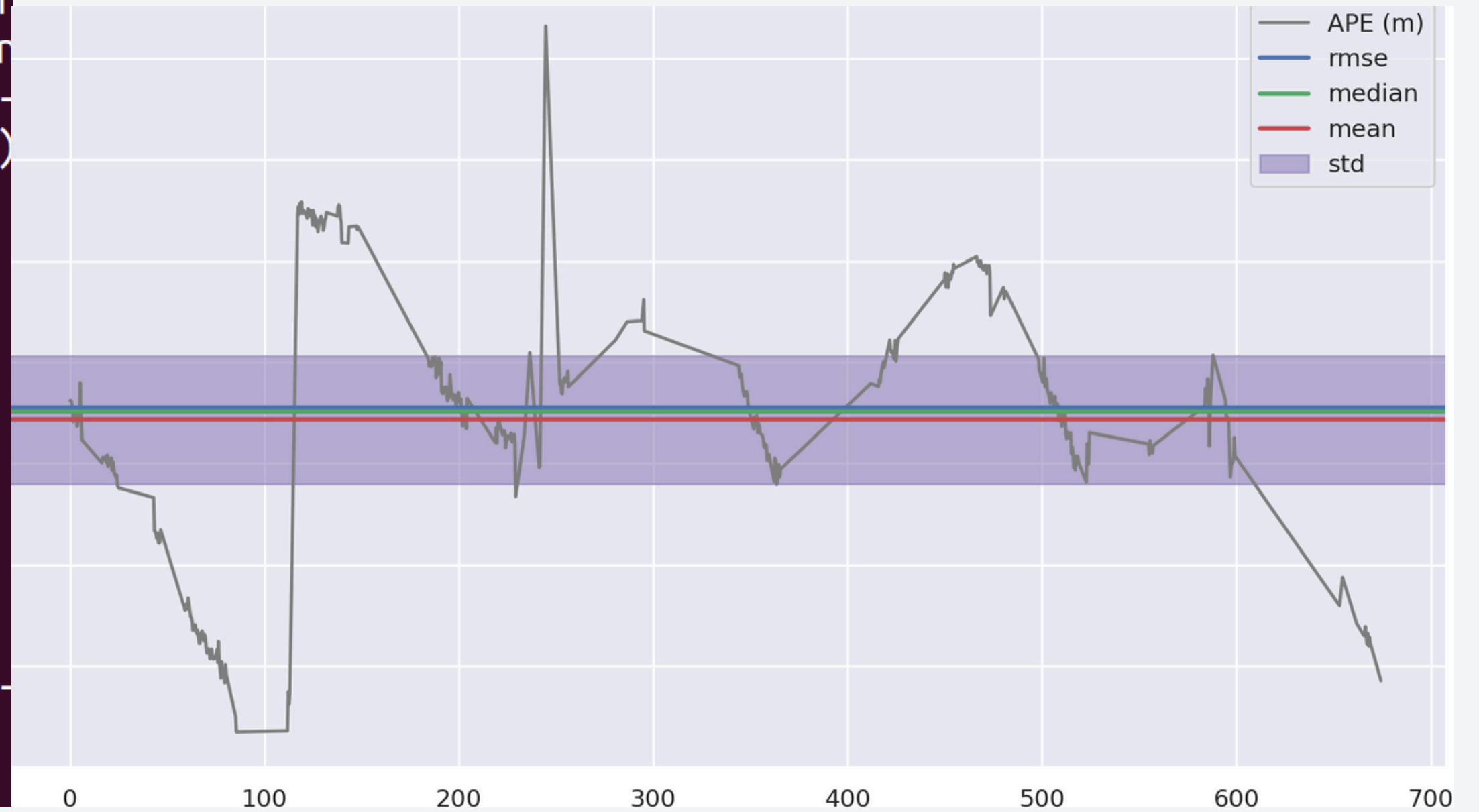
NDT

```
Compared 449 absolute pose pairs  
Calculating APE for translation
```

```
-----  
APE w.r.t. translation part (m)  
(with SE(3) Umeyama alignment)
```

max	2.063302
mean	1.286504
median	1.302228
min	0.670116
rmse	1.311162
sse	771.895877
std	0.253082

```
-----  
Plotting results...
```



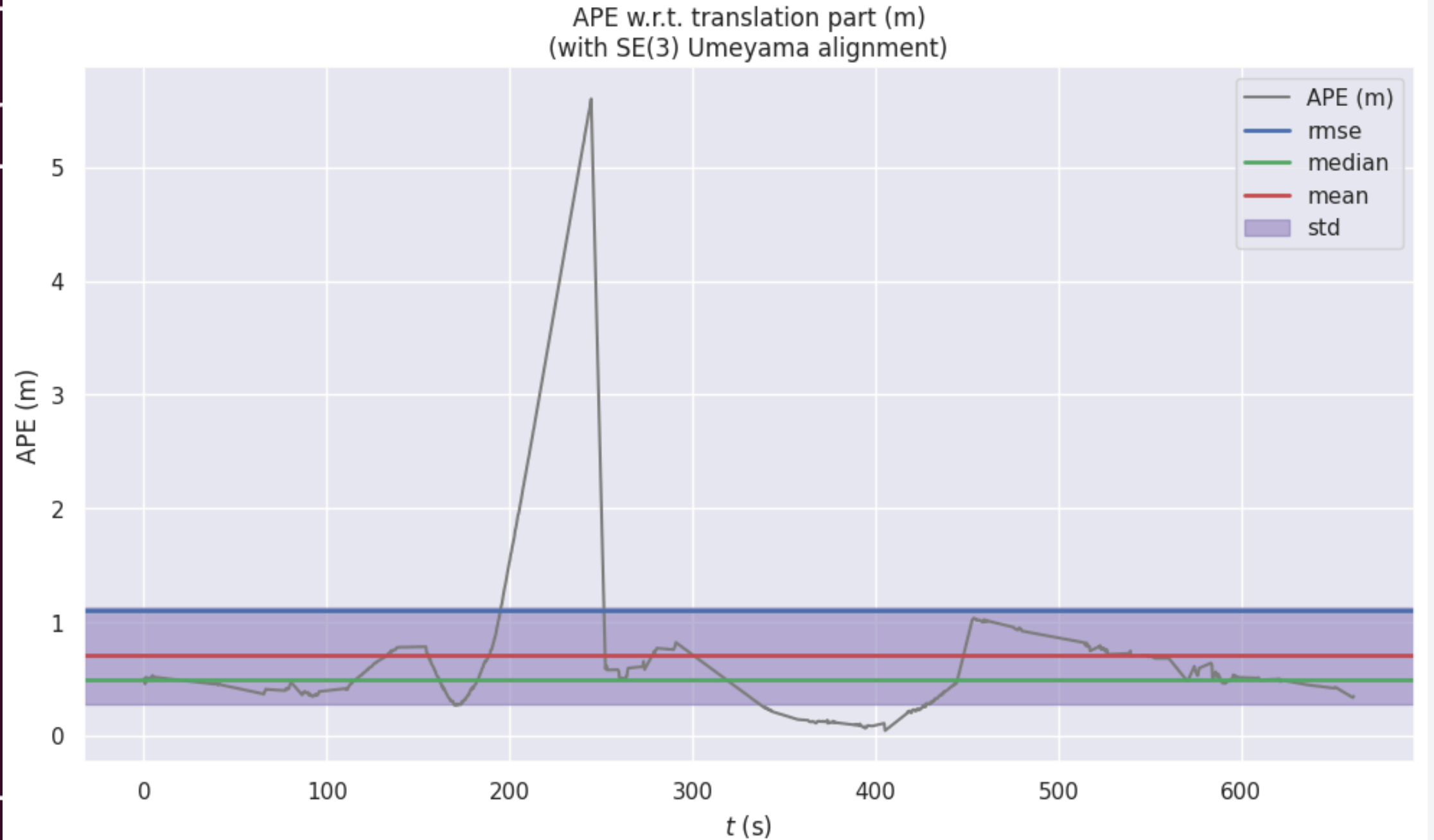
FUSED

compared 561 absolute pose pairs.
calculating APE for translation part

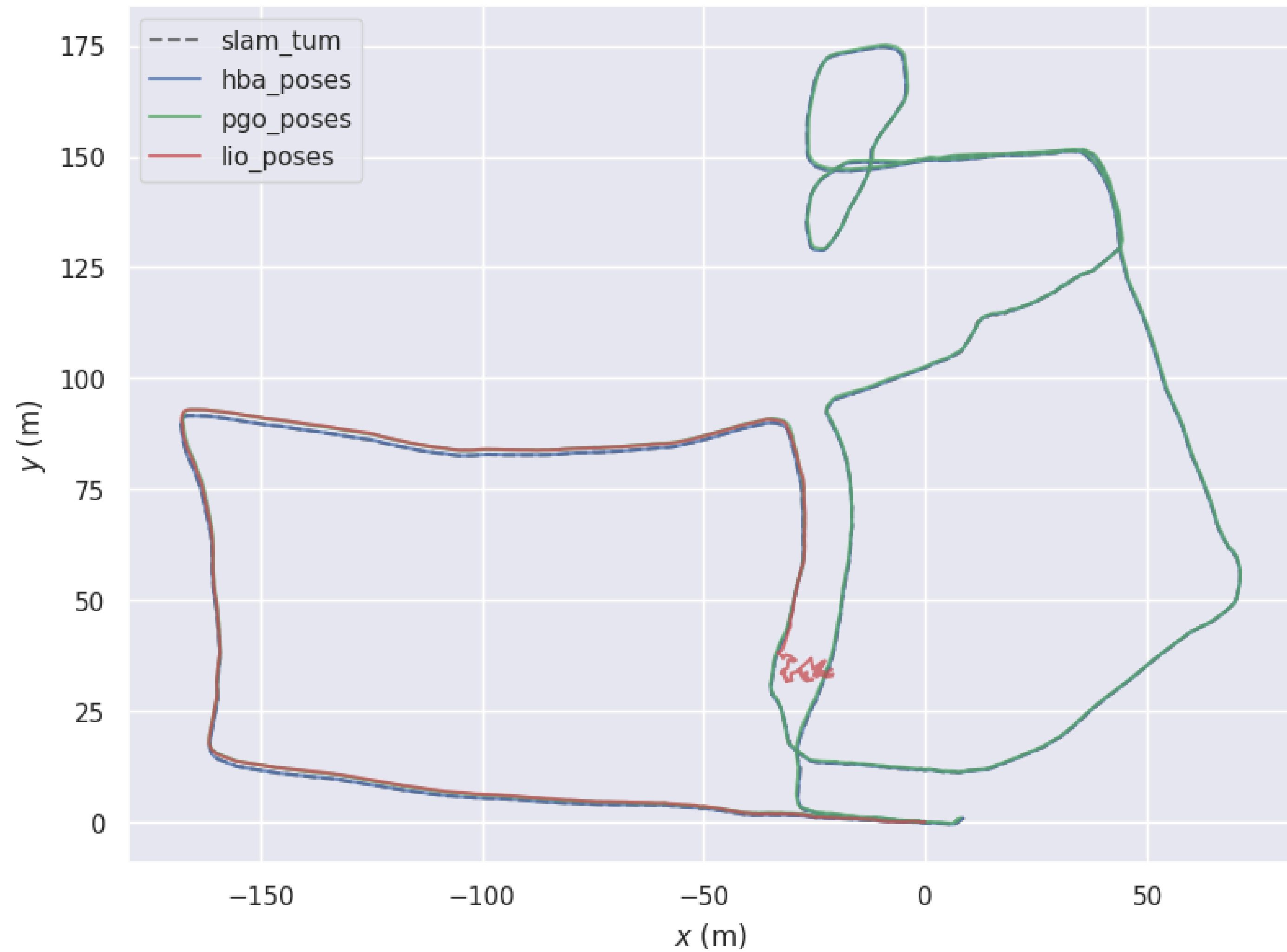
APE w.r.t. translation part (m)
(with SE(3) Umeyama alignment)

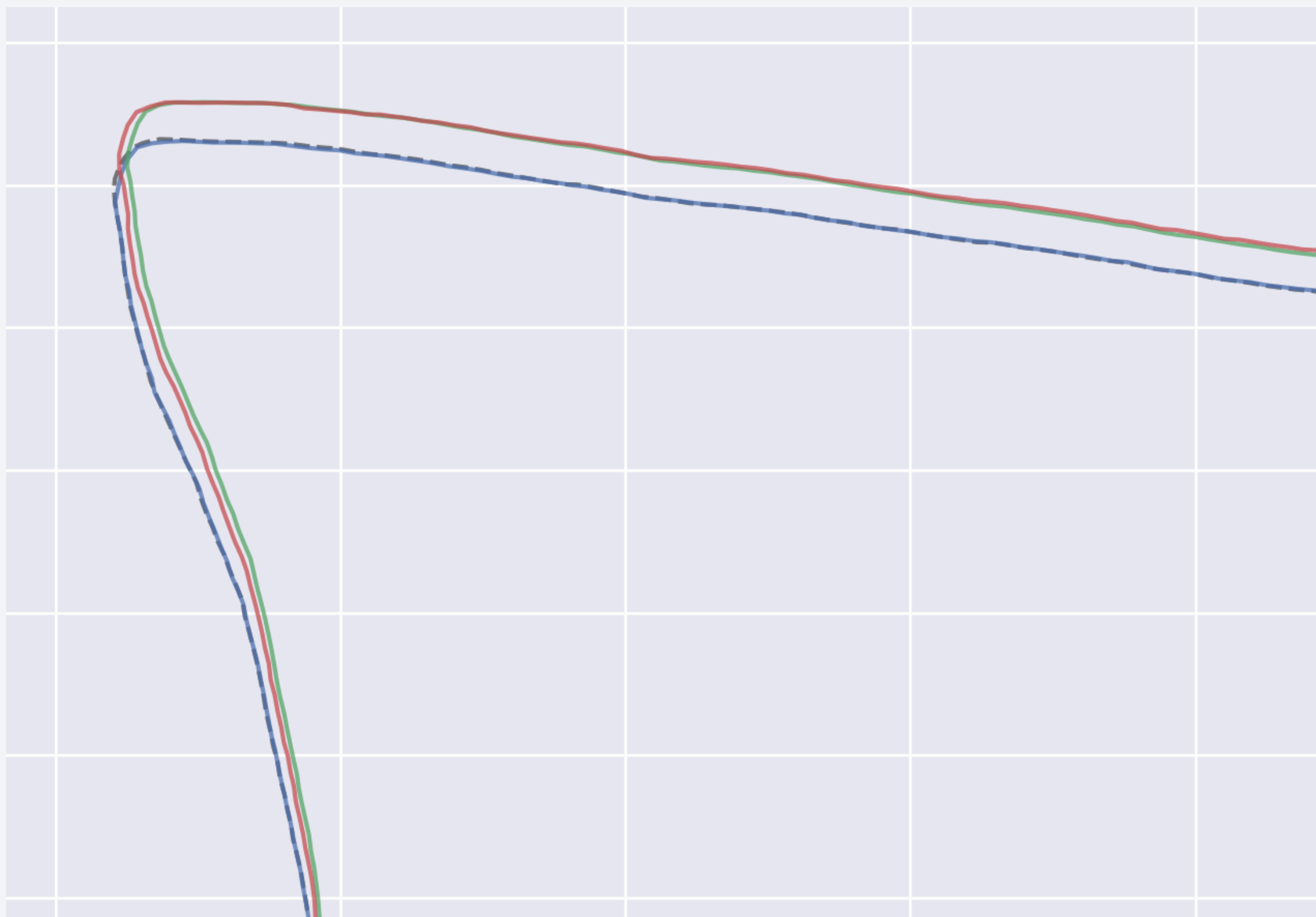
max	5.602368
mean	0.702549
median	0.492793
min	0.042836
rmse	1.101550
sse	680.724611
std	0.848433

Plotting results



TRAJECTORY USING DIFF MAP





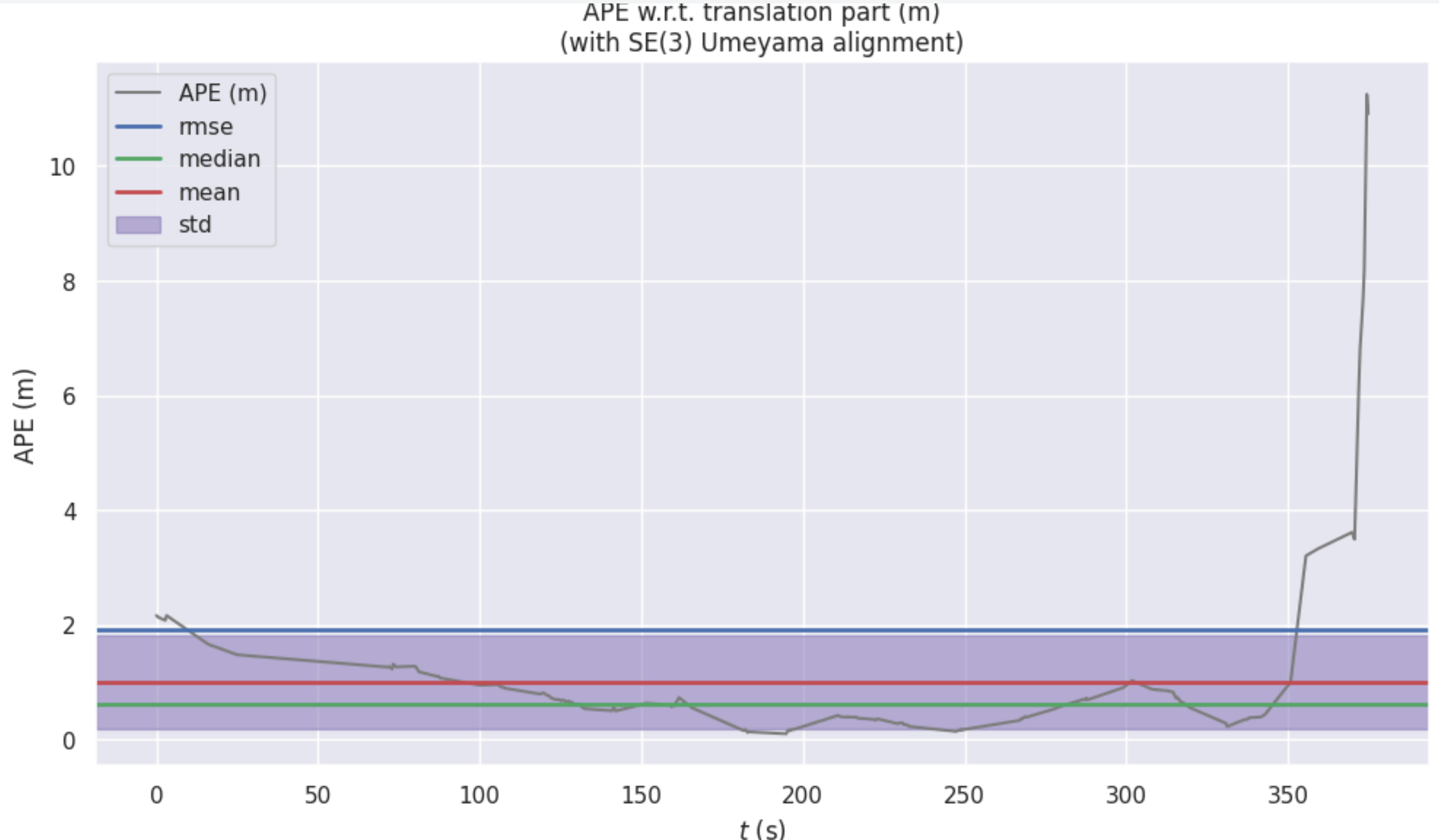
LIO_MAP_ERROR

pared 160 absolute pose pairs
calculating APE for translation

w.r.t. translation part (m)
th SE(3) Umeyama alignment)

max	11.247931
mean	0.998161
median	0.624188
min	0.102214
rmse	1.914769
sse	586.614490
std	1.634018

etting results...



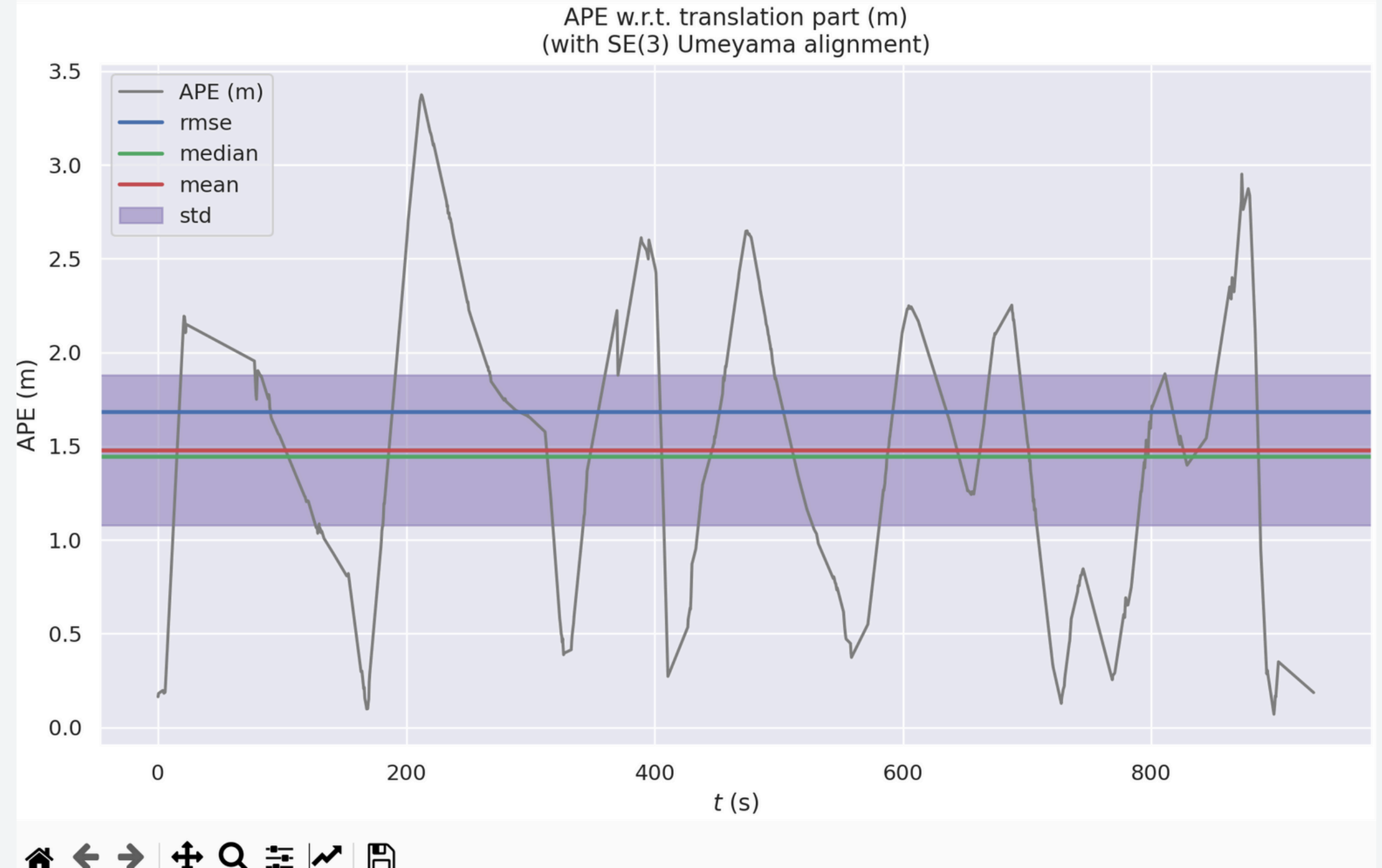
PGO_ERROR

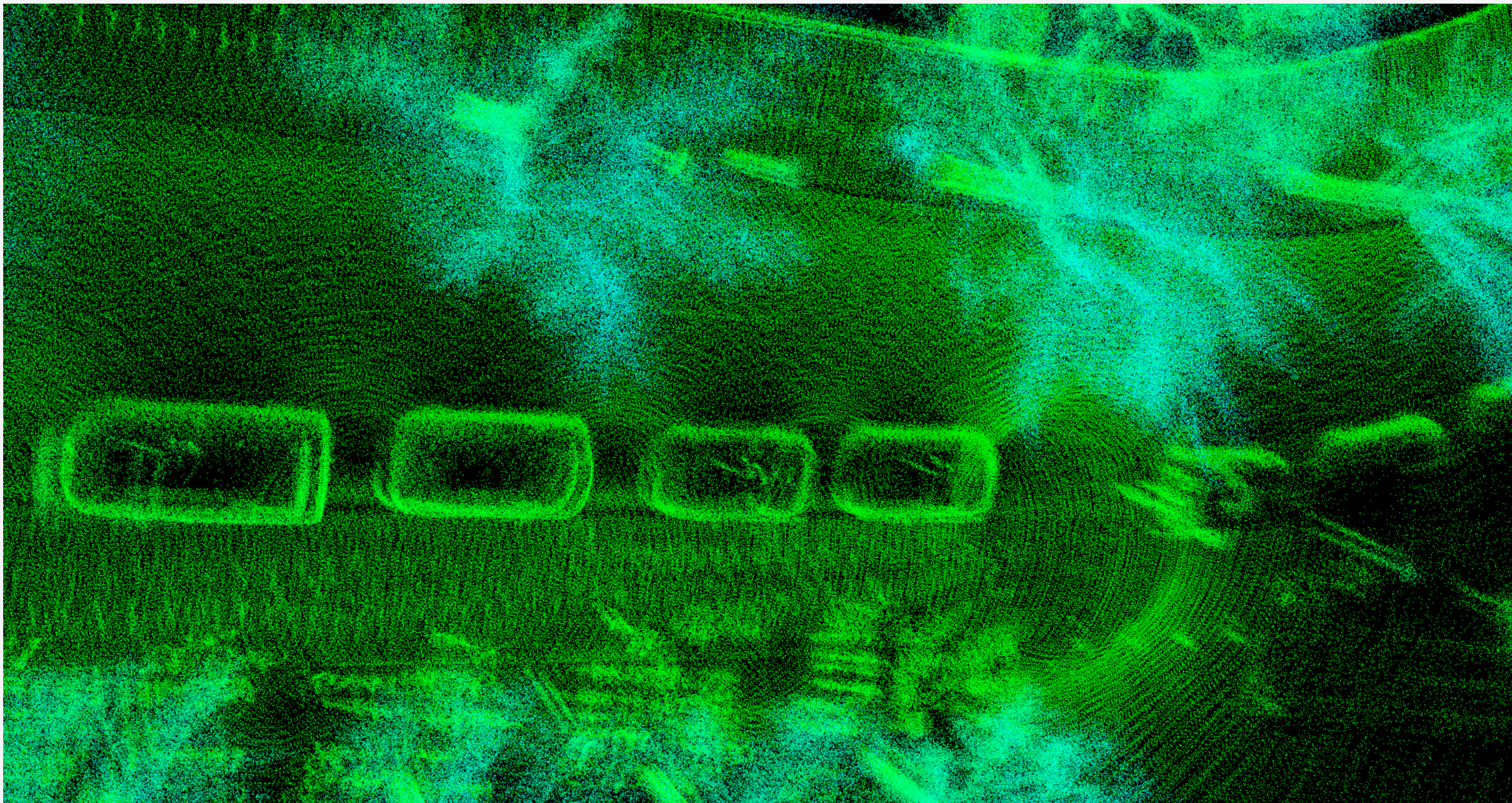
```
-----  
Compared 398 absolute pose pairs.  
Calculating APE for translation pa  
-----
```

```
APE w.r.t. translation part (m)  
(with SE(3) Umeyama alignment)
```

max	3.374560
mean	1.477320
median	1.445145
min	0.069840
rmse	1.680639
sse	1124.169888
std	0.801294

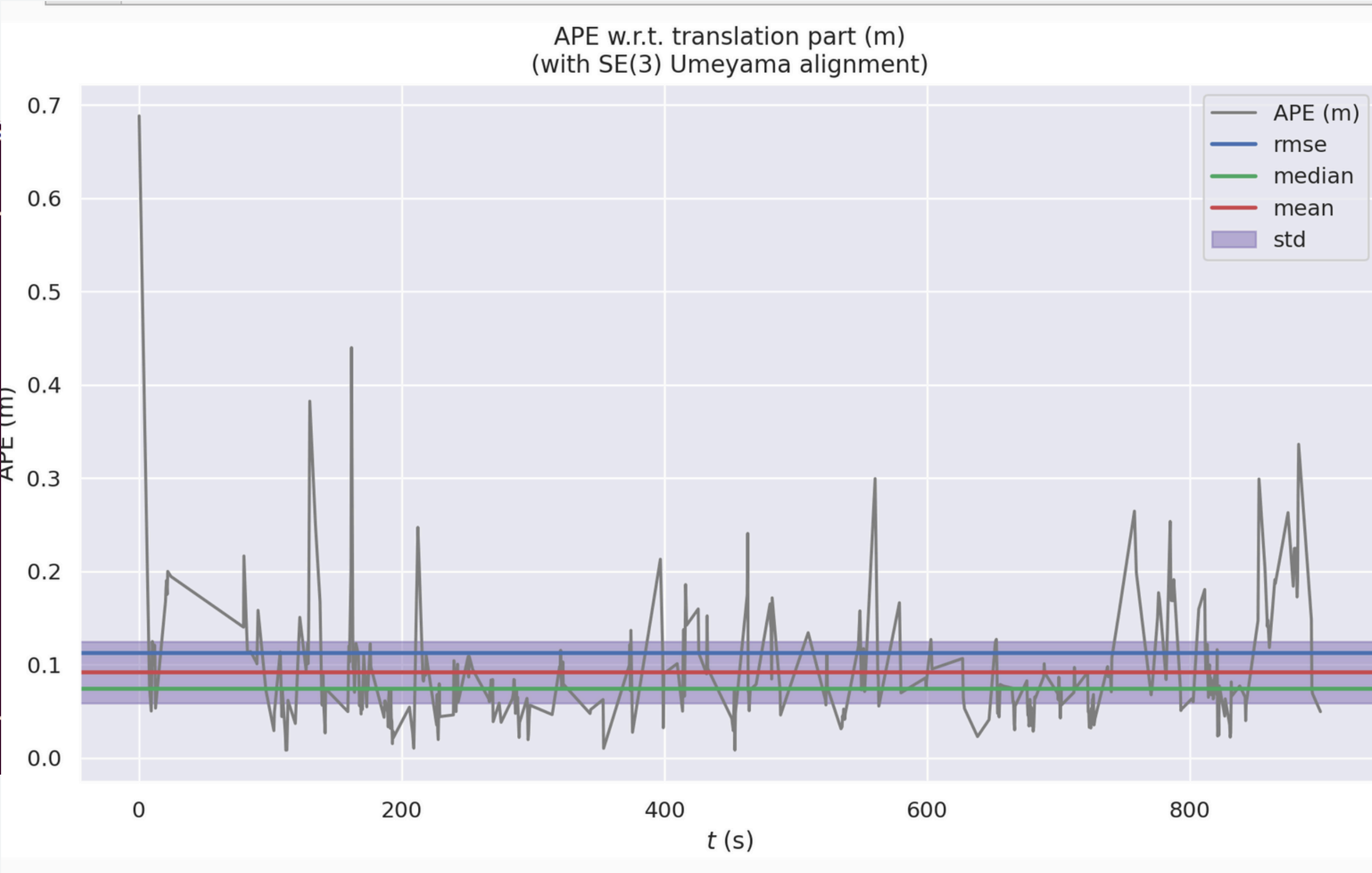
```
-----  
Plotting results...  
█
```



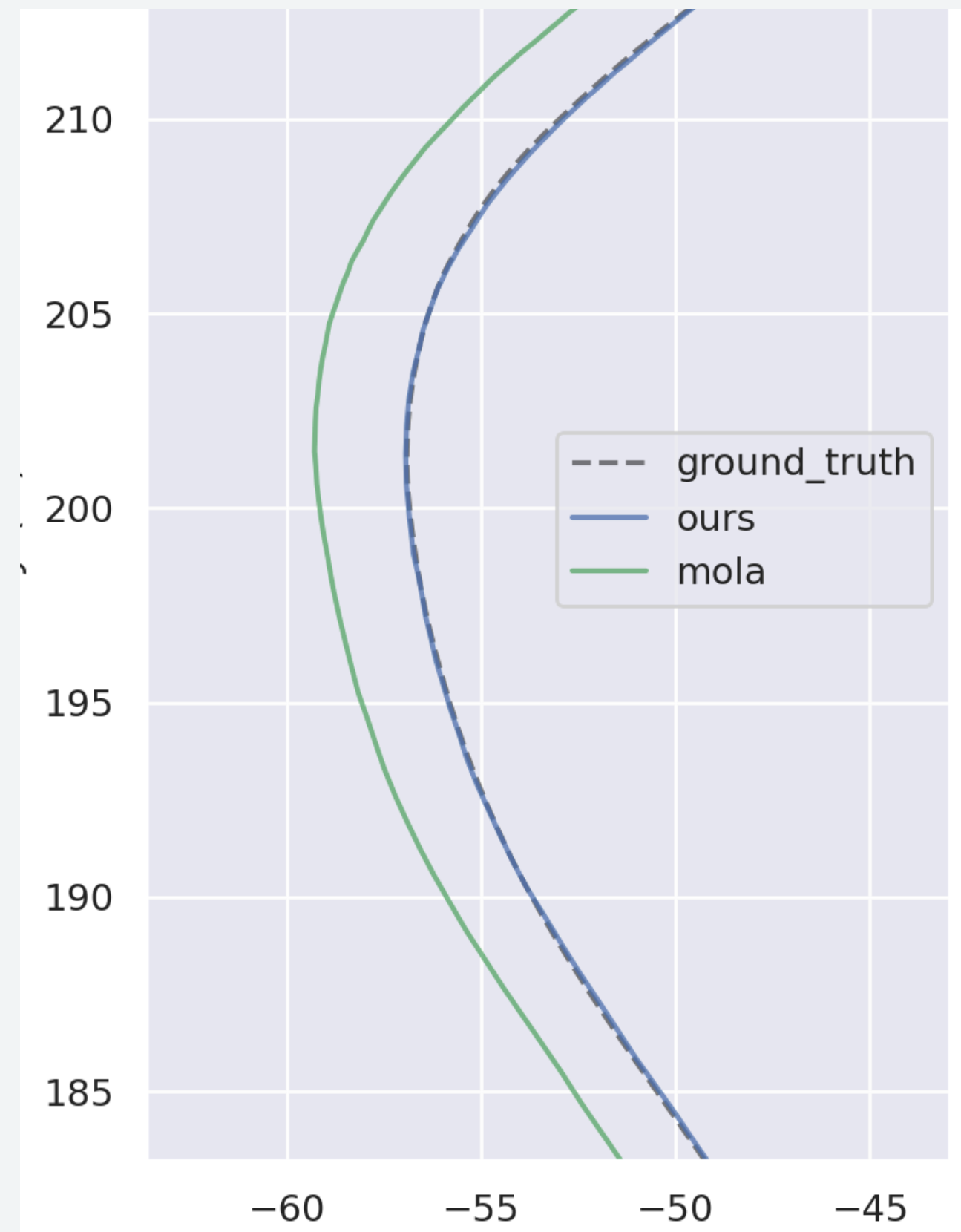
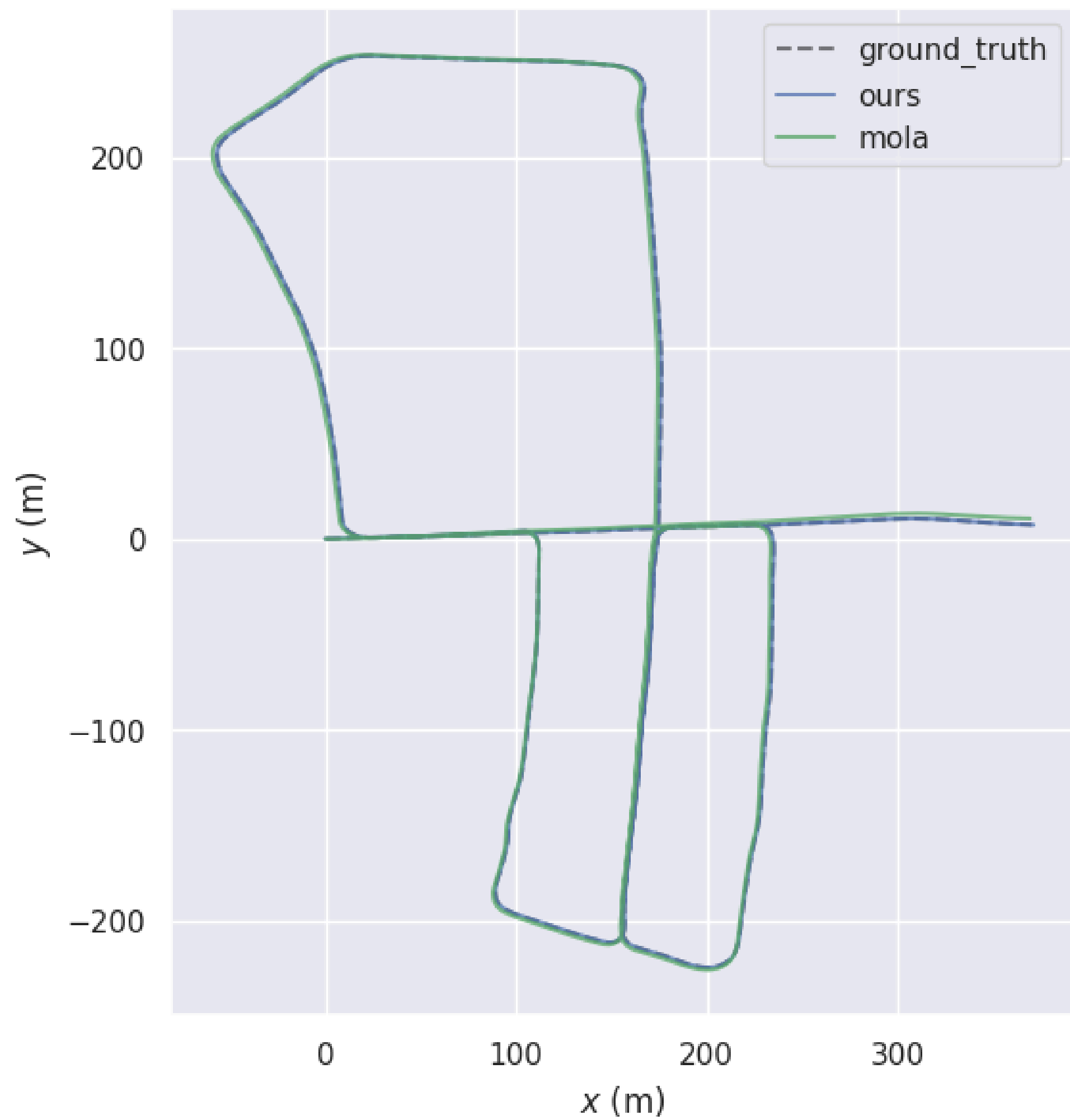


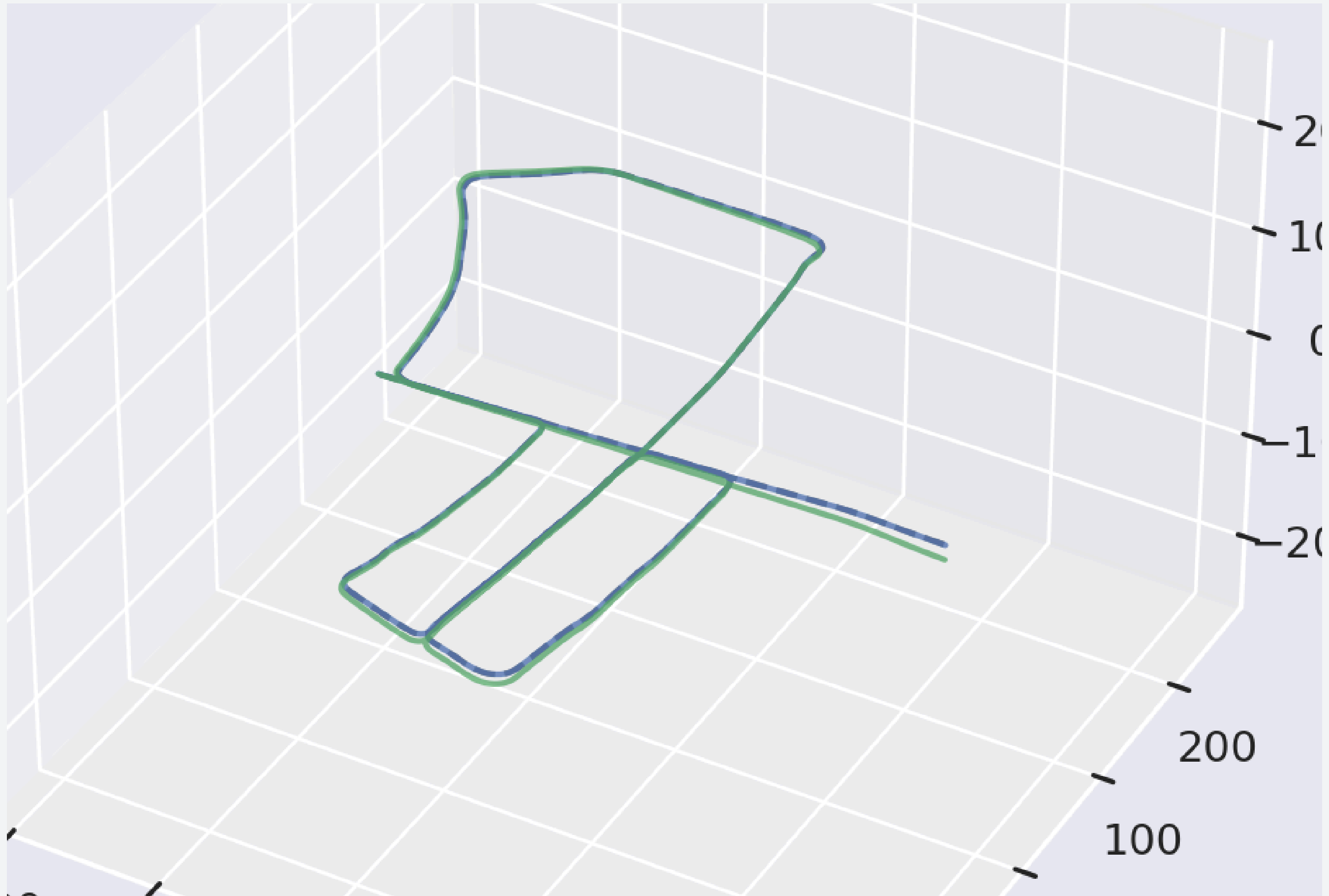
HBA_MAP_ERROR

```
Compared 396 absolute pose pairs  
Calculating APE for translation  
-----  
APE w.r.t. translation part (m)  
(with SE(3) Umeyama alignment)  
  
max      0.688427  
mean     0.091714  
median   0.074451  
min      0.008544  
rmse     0.112779  
sse      5.036735  
std      0.065633  
-----  
Plotting results...
```



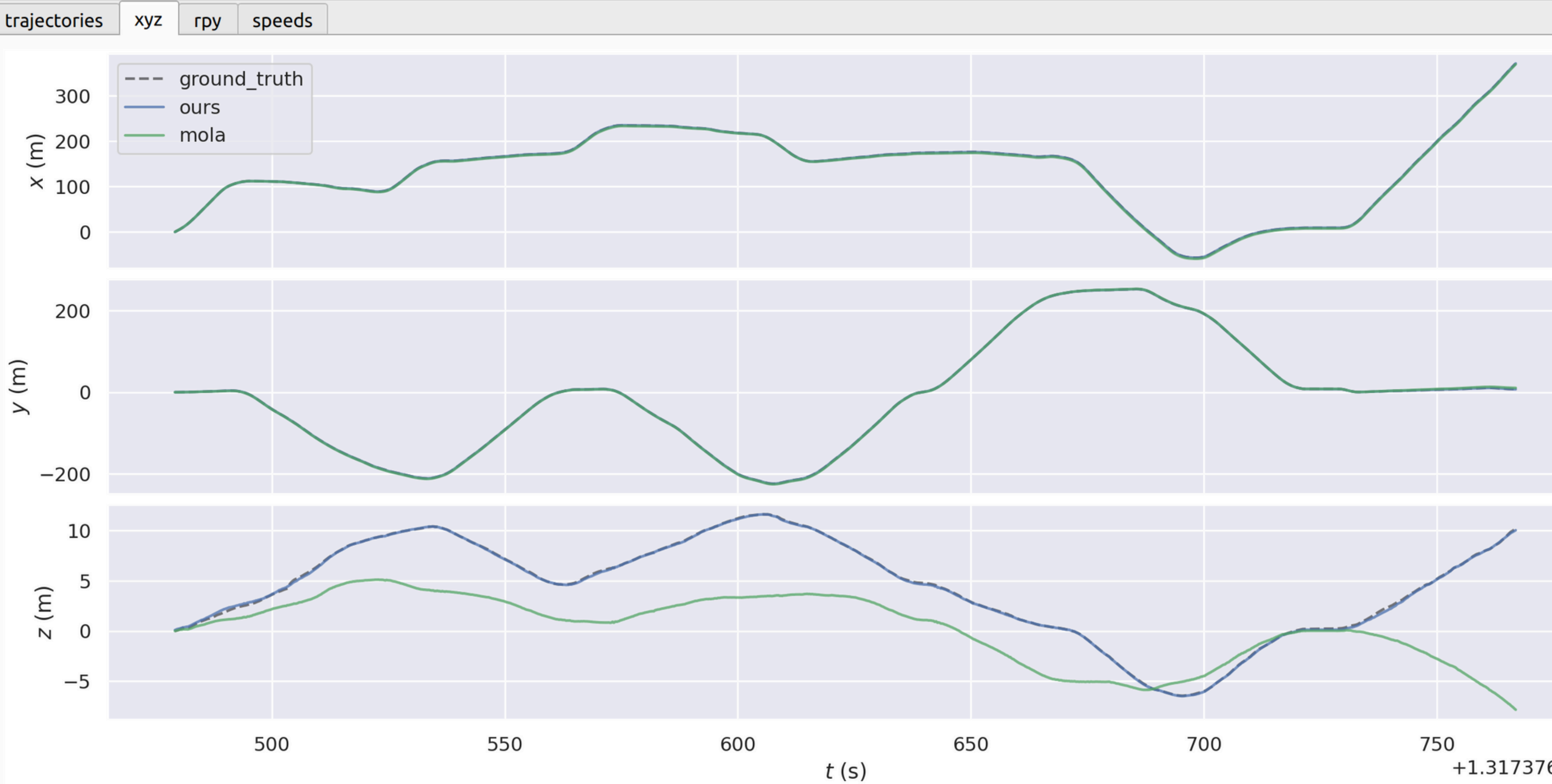
KITITI SEQ 05





KITITI SEQ 05

evo_traj - trajectory plot



OURS

```
Scale correction: 1.0
```

```
-----  
Compared 2457 absolute pose pairs.  
Calculating APE for translation part pose relation.
```

```
-----  
APE w.r.t. translation part (m)  
(with SE(3) Umeyama alignment)
```

max	0.482643
mean	0.120561
median	0.102313
min	0.007258
rmse	0.142932
sse	50.195338
std	0.076777

MOLA

```
[-1.4626088 -0.45434585  0.40876195]  
Scale correction: 1.0
```

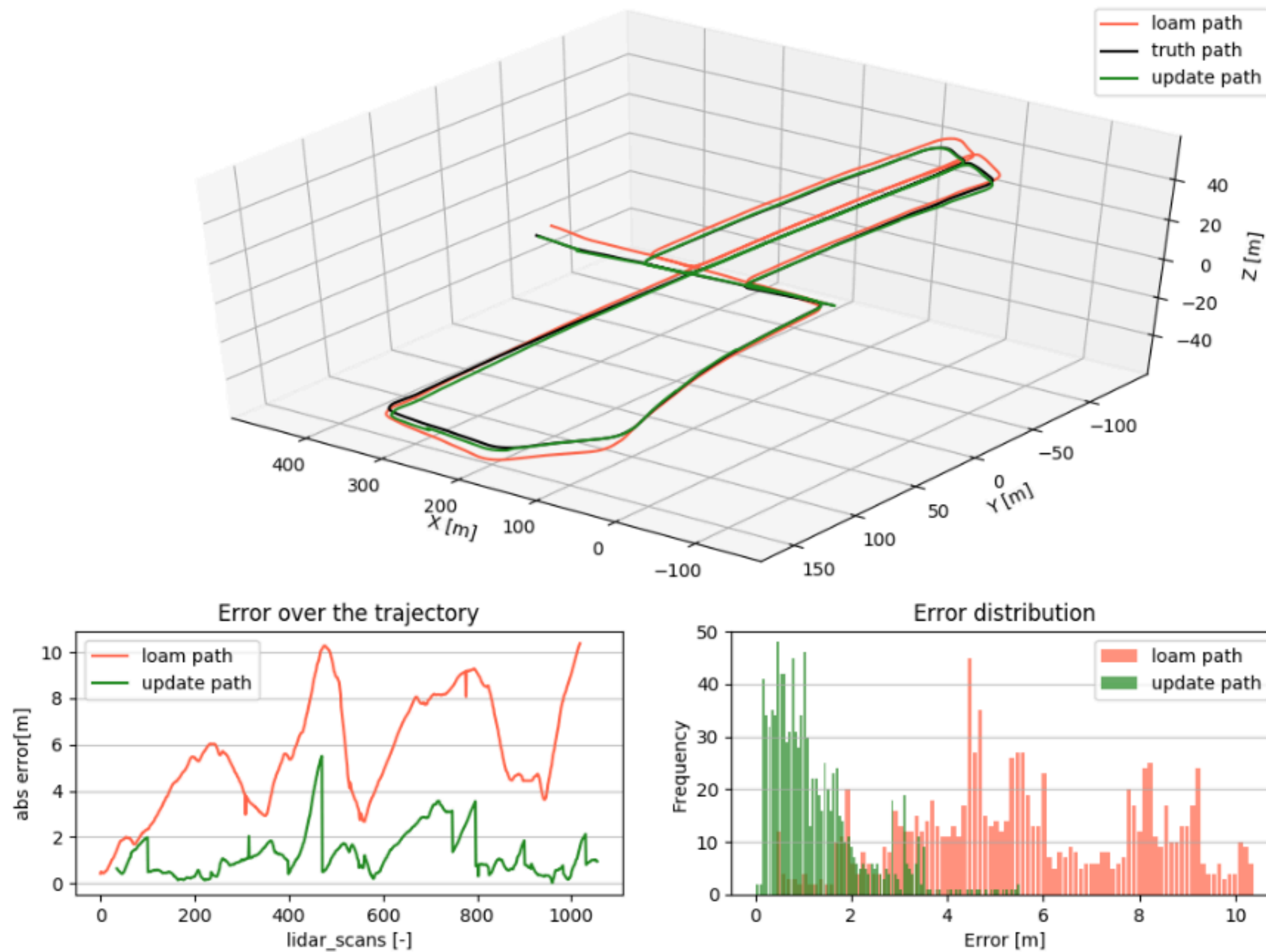
```
-----  
Compared 2761 absolute pose pairs.  
Calculating APE for translation part pose relation.
```

```
-----  
APE w.r.t. translation part (m)  
(with SE(3) Umeyama alignment)
```

max	6.817335
mean	1.538679
median	1.446915
min	0.306204
rmse	1.792923
sse	8875.439191
std	0.920348

```
-----  
Plotting results
```

MAP BASED LOCALIZATION(LOL)



THANK YOU

